Hawkish Fed members fire warning shot across Warsh's bow

The Federal Reserve held rates steady with Stephen Miran again voting for a cut, but three members – while not voting for a hike – signalled a growing sense of unease on the inflation front. This suggests a tougher two-way debate on rates at Kevin Warsh's first FOMC meeting as Chair in June

The Federal Reserve left rates unchanged at Jerome Powell's final meeting as Fed Chair

Fed hawks find their voice

The Federal Reserve has left monetary policy unchanged with officials voting 8-4 in favour of keeping stable rates. Those four are broken down as Stephen Miran again voting for a 25bp rate cut, but with three others – Beth Hammock, Neel Kashkari and Lorie Logan – while not voting for a rate hike, "did not support inclusion of an easing bias in the statement at this time".

The confusing aspect is there isn't specifically a reference to an easing bias. The statement itself says, "The Committee is attentive to the risks to both sides of its dual mandate." This is an unusual way of characterising a vote and it maybe they wanted it included as a shot across the bow of Kevin Warsh, who will be leading the Fed at the next FOMC meeting. They perhaps want to make it clear that they will not be easily swayed to his way of thinking that rates in time can be lowered.

Certainly, the Fed says they will be assessing "a wide range of information, including readings on labour market conditions, inflation pressures and inflation expectations, and financial and international developments". It maybe that the minutes will show they wanted to signal an intention to hike rates if inflation expectations started to push meaningfully higher, risking a more prolonged and broader inflation threat. Markets interpreted it as a hawkish shift, with Fed funds futures contracts discounting stable rates through to year-end, having priced 10bp of cuts in 2026 on Friday.

More disagreement and more market volatility

Today's Senate Banking Committee 13-11 vote in favour of putting Kevin Warsh's nomination to a full Senate ballot means it looks a virtual certainty that he will be signed in as the new Chair just ahead of the 15 May expiration of Jerome Powell's term. The messaging from Warsh is he seemingly wants less of the consensus-driven approach we have seen under previous Chairs, which critics suggest has led to group think that led to slow responses to new events. Instead, Warsh is advocating for an environment where debate and disagreements are encouraged and that looks set to be the case in June. This implies the potential for more surprises and therefore more market volatility. Powell has also said he will continue to serve as a Governor for a "period of time" – his term for that position continues until January 2028 – and this is likely to keep tensions between the President and a more hawkishly positioned Fed, elevated.

Inflation still seen as transitory, but the Fed needs to protect its credibility

Both headline and core inflation have been above the 2% target for five years and looks set to break above 4% next month on gasoline and air fares, so it is understandable why markets and Fed officials are nervous. Officials want to ensure that we don't see higher energy prices feed through into the cost of other goods and services, and so we expect them to talk very tough on this front. This hawkish rhetoric is likely to be stepped up over the next few weeks and until a deal is done in the Middle East that will hopefully provide relief.

Nonetheless, companies have not had much success in passing on the significant cost of tariffs to consumers with CPI goods prices ex-food and energy barely rising 1% year-on-year. Importantly, the current supply shock, focused on fuel prices, is not as broad as the pandemic-related supply chain stresses in 2020/21, and we don't have the same sort of demand impetus that would risk a broader, more persistent inflation story.

Real household disposable incomes are already flatlining amidst a stagnating jobs market, so we see higher fuel prices as being demand destructive via reduced spending power. This is set to weigh on core inflation, and we also need to remember the importance of shelter costs within the US inflation basket (35.5% weighting for headline and 44% for core), which we expect to moderate further. Moreover, if Middle East tensions ease and oil prices drop, there is a high probability of sub-2% year-on-year inflation being achieved in 2027.

Right now, the outlook is highly uncertain and could change very quickly due to developments in the Middle East. We expect the Fed to hold rates steady throughout the summer. Our view, for some time, has been a September rate cut and a December rate cut, but we do recognise that the risk is they are delayed given the current lack of signs of a potential deal. Nonetheless, we still see the prospect of eventual rate cuts being more likely than rate hikes.

Curve flattens on front end scare factor

Earlier, the US 10yr yield had gapped higher, and hit 4.4%. Actually, the whole curve gapped up. The hold-out in the Strait of Hormuz was the catalyst. Some duration selling makes a lot of sense here, where we are in "nowhere land" on a resolution to the war. Extrapolate this, and we could sail back up to the 4.5% area that we hit a few weeks ago. Some of this was echoed in the decision of three Fed members to step away from the underlying easing bias that had dominated policy changes since the wider rate-cutting process began. For these three members, that journey is at an end. The main follow-through from the FOMC outcome is a consolidation at 4.4% for the 10yr, and a further rise in the 2yr to above 3.9% – a flatter curve from the front end. More to come, at least for as long as the hold-off in the Strait continues, and the prognosis there is for the status quo to be in an uncomfortable place.

Between March and April, the implementation notes that accompany the FOMC statement are operationally unchanged, but the tone shifts from active reserve support to maintenance – suggesting the plumbing has calmed enough for the NY Fed Desk to stand down, even as policy stays on hold. Last week the NY Fed cut T-bill buying from US$40bn to US$25bn per month, pointing in the same direction, and indicative of some comfort over the plumbing of the system. The missing ingredient here is the ongoing elevation in the effective funds rate. It remains at 3.64%, a mere 1bp below the rate on reserves at 3.65% (unchanged). It used to be 7bp below (September 2025). The Fed, ideally, would prefer to get it back there. But no big stress here. In all probability, the effective funds rate will be coaxed down as bank reserves slowly rebuild in line with ongoing T-bills buying, even if at a slower pace.

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Why UK bond yields could rise further on a deeper political crisis

Britain's local elections are a potential market flashpoint. Minimal political risk is priced, even as bond yields hit post 2008 highs of 5%. Borrowing costs could initially climb if the political situation worsens, but a leadership change need not materially alter the fiscal or Bank of England outlook

UK Prime Minister Keir Starmer is in a fight for political survival

Local elections: What's happening and why it matters

Britain's local elections come at a particularly sensitive moment for UK financial markets. Government bond – gilt – yields have just risen above 5% for the first time since 2008, a time when the economy was growing significantly faster than it is today. Much of the latest spike is down to inflation, but the politics can't be ignored. Prime Minister Keir Starmer is in a fight for political survival and a bad election night on 7 May could prove fatal.

Up for grabs are 5013 local council seats – a little over a quarter of Great Britain's total – plus the Scottish and Welsh parliaments. Starmer's Labour Party looks set to lose up to 90% of the 2268 of its council seats up for election, according to some predictions.

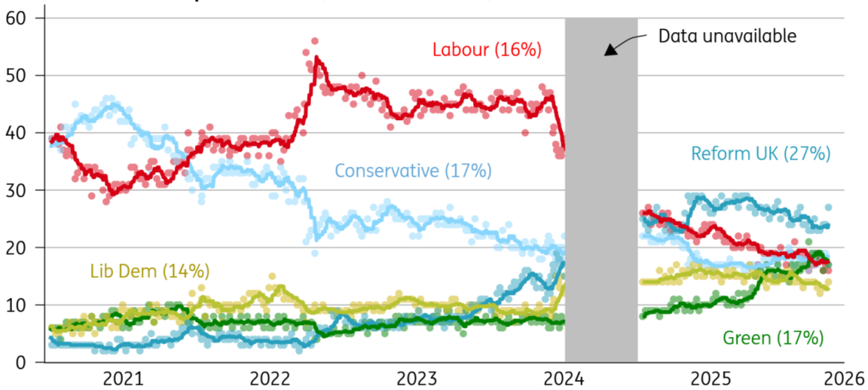
The scale of the defeat will be key. But so too is what drives it.

We know that Nigel Farage's Reform UK is likely to be the main beneficiary of May's elections. The party is running between 25-30% in national polls, up from 15% at the 2024 election. Much of that gain has come at the expense of the Conservative Party, but Reform's rise still poses an issue for Labour.

In Britain's First Past the Post electoral system, general elections are typically won or lost depending on whether the left or the right is more divided. And Reform UK's increasing dominance of the political right comes as the left is increasingly split.

The rise of the Green Party is a bigger issue for Labour than Reform UK

UK General Election poll - YouGov (latest in brackets)



Source: YouGov

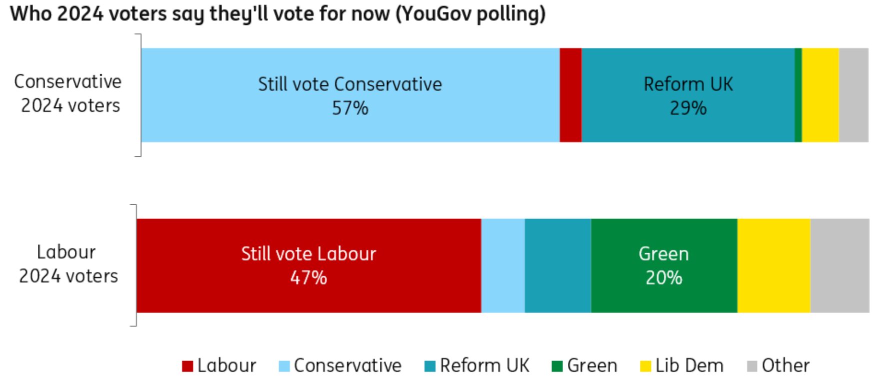
Solid line is the one month moving average

Labour, the Lib Dems and the Green Party are all polling between 14-17% nationally. That's a collapse from 34% at the last election for Labour, but crucially a surge for the Greens, whose share has doubled from 7% in June 2024. If a general election took place tomorrow, Reform would be much more likely to command a majority in Parliament.

To illustrate the point, at 14% of the national vote, Reform won just five seats at the 2024 election. At 24%, that would surge to 188 seats; at 31%, it's 335, according to Multi-level Regression and Post-stratification (MRP) polling by Electoral calculus in January and April of this year. 326 seats are needed for an outright majority.

Britain isn't on the brink of a general election. The next vote isn't until 2029. But Labour's fortunes in three years' time rely on stemming the tide of votes to the Greens and other left-wing parties. A big defeat at the local elections would inevitably focus minds within Labour on what it needs to do to win back support. Polymarket puts a 67% chance on Starmer leaving office by year-end.

20% of Labour 2024 voters would now vote Green



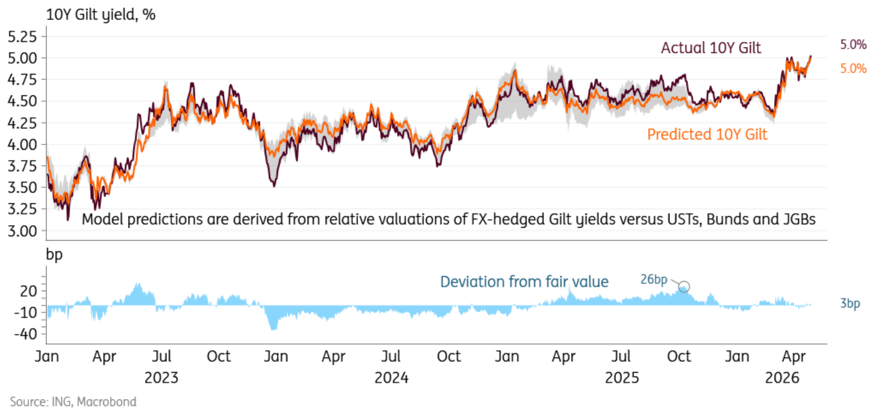
Source: YouGov

How bad would a leadership contest be for markets?

Gilts are already under scrutiny due to inflation risks, and adding political uncertainty to the mix could further push (global) investors to look elsewhere. Whilst we've seen some swings in gilt yields on days with political headlines, we estimate the risk premium on 10Y gilts is still limited. If the situation were to escalate or a leadership contest were to be triggered, the 10Y yield could easily rise another 10-20bp, which would be in line with the nervousness seen in markets prior to the budget announcement last year.

The chain of logic among investors looks like this: a new Labour leader and prime minister would likely mean a new chancellor. A new chancellor, under pressure to take the party towards the left to counter the threat posed by the Green Party, might be more pro-spending. That potentially means new, looser fiscal rules. And new fiscal rules would likely mean more borrowing.

There's virtually no risk premium in 10-year gilts



Source: Macrobond, ING

But the reality might be more nuanced. The infamous 2022 "mini budget", which sparked a sharp though relatively short-lived spike in yields – and mortgage rates – is well etched into the political memory in Westminster. No politician wants to risk triggering a re-run.

Sure, leadership hopefuls may criticise the Office for Budget Responsibility, the independent arbiter of Britain's fiscal rules. Angela Rayner, the betting markets' favourite to become the next Labour leader, has said the OBR fails to account for the wider benefits of public investment.

But that is not the same thing as proposing a new set of fiscal rules. Leadership contenders will be under heavy pressure to rule out changes.

Remember, too, that a leadership contest can only be triggered if 20% of Labour MPs (that's 81) back a single rival candidate. Labour lawmakers might ultimately agree on the need for a new leader, but will they back the same horse?

It might not come to that. Leaders often go when their cabinet turns against them; we saw that with the departure of Boris Johnson in 2022. Nevertheless, a successful leadership candidate will need to command support from across the party. And that might require them to pick a chancellor from a different faction.

A leadership contender on the left of the party might pick someone from its right to head up the Treasury as the price of gaining sufficient support. Doing so might also help calm investor nerves about an impending surge in borrowing.

There also don't appear to be loud calls from within the party for big fiscal loosening right now. A recent speech by two Labour MPs, representing opposite wings of the party, pitched a new economic agenda focused on supply-side policies like tax reform, over big increases in public spending. There haven't so far been widespread calls for a large universal energy support package, either.

Political noise doesn't necessarily boost chances of BoE hikes

The bottom line is that the near-term fiscal trajectory – borrowing over the next one or two years – is unlikely to look wildly different whoever is in charge. And notwithstanding the war in Iran, the fiscal backdrop had been looking brighter; the structural deficit was – and still could – fall this year on the ongoing freeze in tax thresholds.

That matters because it suggests domestic politics is unlikely to have a large bearing on Bank of England policy. And even if it does, it's unlikely the next budget will come earlier than currently planned in late autumn. The Bank can't act on fiscal policy until it is formally announced. The overwhelming focus will remain on the energy crisis and its impact on inflation.

Longer-term, the story becomes more problematic for markets – but again, we suspect this is true whether it is Starmer or someone else in Downing Street. The main fiscal rule – to bring day-to-day spending into balance with tax revenues within three years – is currently met on paper. But only by promising sizeable tax rises, many of which aren't scheduled to kick in until 2029. Real per-capita spending on public services is also slated to fall in that year – a year, don't forget, when the next general election is likely to fall.

In an environment where Labour is struggling in the polls, there will be a lot of pressure to either cancel or offset those plans. And that's before considering the likelihood that defence spending plans will need to be ramped up.

FX markets price little UK election risk

In FX markets, it feels like there is fatigue over the event risk of these local elections. Sterling bears have been talking about 7 May for a couple of months now, but these risks have been swamped by developments in the Middle East. Looking at the FX options market, the volatility term curve shows just a minor kink around the local election date. And for FX options expiring around the day of the event risk, a 30 pip range is priced for EUR/GBP compared to 14 pips on a 'normal' day.

As above then, a fresh bout of risk premia coming through the fiscal side could demand some temporary weakening in the pound. Until, however, energy markets calm down, it is hard to see EUR/GBP breaking clear of swap differentials. These have narrowed in sterling's favour based on the view that the BoE is going to react more aggressively (tighten) than the ECB to higher energy prices. Until the BoE can forcefully get across the view that it is going to hike less than the ECB, it seems as though EUR/GBP can linger a lot longer in the middle of a 0.86-0.88 range than most – including us – had been expecting.

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Rates Spark: Big rate decisions as oil tests highs

Despite oil prices testing new highs, markets are not pressuring the European Central Bank or Bank of England to hike this meeting, but the stakes are higher for June. The return of a UK political risk premium could push 10Y gilt yields well above 5%. The Fed set the tone by leaving rates unchanged, but the tone has changed, with three dissents to an easing

We expect the European Central Bank and Bank of England to keep rates on hold

US curve flattens on front end scare factor

Earlier Wednesday, the US 10yr yield had gapped higher, and hit 4.4%. Actually, the whole curve gapped up. The hold-out in the Strait of Hormuz was the catalyst. Some duration selling makes a lot of sense here, where we are in "nowhere land" on a resolution to the war. Extrapolate this, and we could sail back up to the 4.5% area that we hit a few weeks ago.

Some of this was echoed in the decision of three Fed members to step away from the underlying easing bias that had dominated policy changes since the wider rate-cutting process began. For these three members, that journey is at an end. The main follow-through from the FOMC outcome is a consolidation at 4.4% for the 10yr, and a further rise in the 2yr to above 3.9% – a flatter curve.

More to come, at least for as long as the hold-off in the Strait continues, and the prognosis there is for the status quo to be in an uncomfortable place.

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The missing piece here is the ongoing elevation in the effective funds rate. It remains at 3.64%, a mere 1bp below the rate on reserves at 3.65% (unchanged). It used to be 7bp below (September 2025). The Fed, ideally, would prefer to get it back there. But no big stress here. In all probability, the effective funds rate will be coaxed down as bank reserves slowly rebuild in line with ongoing T-bills buying, even if at a slower pace.

No early ECB hike anticipated as stagflationary pressures complicate the outlook

When the ECB meets on Thursday, it faces the dilemma of further oil price rises stoking inflation expectations while at the same time there are more signs of mounting macro headwinds.

On the back of Brent oil climbing above US$115/bbl, money markets are fully pricing in three hikes from the ECB by the end of the year. But with a view to this meeting, market expectations remain relatively muted, with just 3bp priced into the forward, implying a probability of somewhat more than 10% for a hike.

It is a tail risk, but it would run counter to the official communication ahead of the blackout period, which suggested that April would likely be too early to decide on a hike given the limited availability of new data. Delivering a surprise rate hike at this stage would probably be seen more as a front loading than a change of the ECB reaction function, and long-end rates might see limited pass-through resulting in a bear flattening.

We believe the ECB could resort to a more hawkish tone to underpin its inflation-fighting credibility, but refrain from immediate action given that the overall geopolitical situation remains very uncertain. At the same time, the increasing signs of adverse growth effects will make signalling an aggressive hiking cycle less straightforward.

Political risks add to hawkish Bank of England pricing

We also expect the Bank of England to remain on hold, but markets price in 80% of a 25bp rate hike by June, which seems on the hawkish side. One could argue that UK inflation never made it back to target, thereby complicating the outlook. But similarly to the eurozone, the economic demand backdrop is weaker than in 2022, and the still relatively limited impact on gas prices should help mitigate second-round price effects. For this reason, our economist thinks the bar for a hike remains high, and this resonates with the earlier pushback from Governor Bailey against the hawkish positioning of markets.

Meanwhile, 10Y gilt yields are closing at new highs well above the 5% handle, and with political risks looming, there could be more risks to the upside. The push higher of sterling rates has thus far been driven by the sharp repricing in inflation expectations, more so than we've seen in the eurozone or US. But with Starmer's position as prime minister looking more fragile, we could see the build up of a political risk premium forcing gilt yields even higher. We estimate that the 10Y gilt yield may rise 10-20bp if investor sentiment turns wearier as it did during the budget announcement last year.

Thursday's events and market view

The policy-setting meetings of the ECB and BoE will take the spotlight, but there are also relevant data releases scheduled for the day.

The eurozone releases the flash April CPI figures. The market looks for a headline increase to 3% year-on-year, although country releases so far point to a slightly lower print. Core inflation is expected to come in at 2.2% YoY versus 2.3% in March. The eurozone will also release the advance estimate for first-quarter 2026 GDP growth and the unemployment rate for March.

In the US, the PCE inflation for March is expected to rise to 3.5% in the headline. We will also get the personal income and spending data for that month, the weekly jobless claims numbers and also advanced readings for first-quarter GDP growth.

No bond issuance is scheduled for the day. In other news, the EU raised its bond funding target for 1H26 from €90bn to €100bn following the final adoption of the €90bn Ukraine support loan. This will be accommodated by "marginal increases" of the already pre-announced issuances. The funding target for the entire year was raised from €160bn to €180bn.

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A dovish-leaning hold by the Bank of Canada

The Bank of Canada held rates steady at 2.25%, but sounded relatively relaxed for now on inflation risks. At the same, there are lingering concerns about the jobs market and USMCA renegotiations. We share those concerns, and in our baseline scenario expect no hike this year, alongside only a gradual decline in USD/CAD over the remainder of this year

Bank of Canada Governor, Tiff Macklem

Jobs and USMCA concerns keep BoC slightly dovish

The Bank of Canada left its policy rate at 2.25% as expected, but the overall tone of the accompanying statement leant somewhat dovish relative to market expectations. The BoC acknowledged the impact higher energy prices were having on headline inflation and inflation expectations, and Governor Tiff Macklem said that a hike might be needed if oil prices were to remain elevated for a long period. However, the Bank also emphasised that core inflation has behaved well and that the proportion of components of CPI that are above target has declined.

Meanwhile, the jobs market is still described as "soft" with weak hiring and job losses seen in sectors exposed to US tariffs. Growth projections also remain modest, with 1.2% GDP growth predicted for this year, 1.6% next, and 1.7% in 2028. Those 2027 and 2028 forecasts are a tenth of a percentage point below the consensus growth estimates.

While higher energy prices are a boon for Canada's economy given it is a major net producer of oil and gas, the ongoing uncertainty about trade policy continues to hold back sentiment. The coming evaluations of the USMCA trade deal between Canada, the US and Mexico and the potential for the US to put more of a squeeze on trade partners prompted Governor Macklem to warn that if there are significant rule changes that harm Canada, then a rate cut may be required.

Our call is still for no hikes

As such, the BoC seems to be hedging its bets a little. For now, the Bank is prepared to "look through the war's immediate impact on inflation". Assuming oil prices come down, and US tariffs remain unchanged, "a policy rate close to current settings looks appropriate". The upside risk stems from a prolonged period of elevated inflation costs that starts to feed through into broader price increases, while the downside risk is increased trade restrictions with the US.

In terms of our view, our base case is that the Middle East situation will gradually ease, we see a slow resumption of tanker flows through the Strait of Hormuz over the next couple of months, and energy prices do indeed start to come down. We also see the potential for a tough period of talks over the USMCA deal that will leave a cloud of uncertainty over the economy and weigh on the jobs market. In this situation, we continue to favour stable policy rates through to the end of the year.

Dovish BoC restrains CAD

USD/CAD briefly traded on the strong side after today's BoC announcement, although market pricing didn't shift meaningfully. Expectations remain for a 25bp rate hike at the October meeting, so more hawkish than our baseline call.

Today's relatively dovish BoC endorses our end 2Q call for 1.37 in USD/CAD. We do see some downside risks to that forecast given resilience in equity markets and oil prices remaining more elevated than longer. But for now, we are assuming there will be a period of elevated uncertainty around USMCA renegotiations in June, leading to upward volatility in the pair.

We expect better CAD performance in 2H26, when a new USMCA deal could ultimately be agreed and the Federal Reserve should cut rates. We expect 1.34 by year-end in USD/CAD, with risks to the downside.

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Disclaimer

US solar set to see a significant supply chain reconfiguration

The reshoring of US solar manufacturing is moving forward, but only gradually. Continued reliance on China, combined with tariffs and FEOC rules, means developers face significant work to map, verify, and potentially reconfigure their sourcing strategies

Solar reshoring across the value chain is still progressing slowly, though momentum is building

Steep tariffs and strict foreign entities of concern (FEOC) rules under the Trump administration are accelerating a reset of the solar supply chain – on a compressed timeline. This poses challenges for developers that have long relied on China's dominance in global production of cheap solar and battery equipment. Components with minimal Chinese exposure are likely to be in high demand, pushing up prices.

As developers rework their supply chains, short-term fixes may not be enough. More durable solutions will require building and maintaining trade relationships that can withstand future policy shifts.

Over time, a vertically integrated domestic supply chain is achievable, though it will require incentives and trade protections. The significant volume of solar cell capacity currently under construction in the US suggests progress in the right direction.

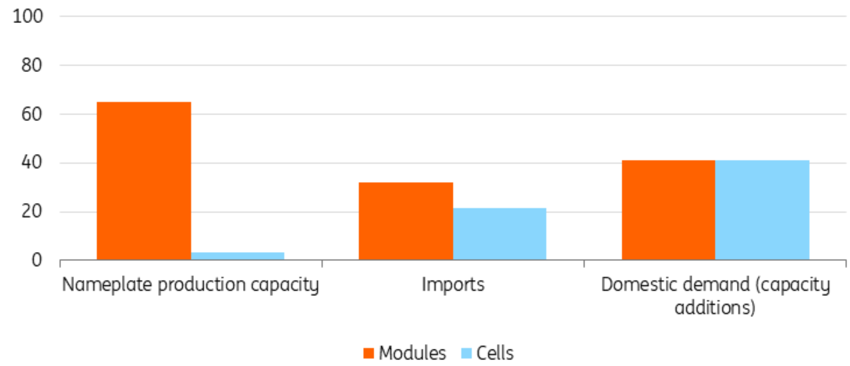
Seemingly high US module capacity does not ensure self-sufficiency

US solar manufacturing capacity has expanded rapidly. The Solar Energy Industries Association (SEIA) estimates that as of early 2026, US solar module manufacturing capacity increased 50% annually to 65GW, much higher than the domestic demand of roughly 40 GW in 2025.

While we think this is a credible and impressive data point, we caution against the interpretation that the US is self-sufficient on solar modules. In fact, the US still imported 32GW of modules last year, despite having 65GW of module capacity targeting mostly domestic customers.

US solar manufacturing capacity, imports and demand

GW

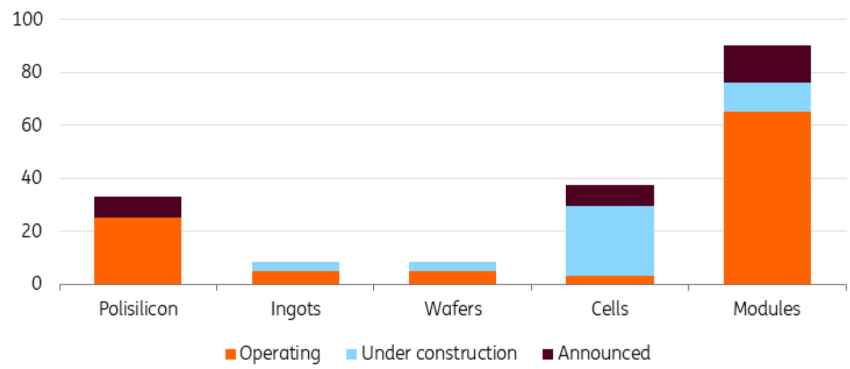


Source: ING Research, Solar Energy Industries Association, BNEF, Wood Mackenzie

One main reason behind this is upstream manufacturing bottlenecks. Further up the supply chain of crystalline silicon (c Si) modules, cell capacity is just 3.2 GW, while wafer and ingot capacity are each around 5 GW. Therefore, US module capacity is not true 'manufacturing capacity', but largely '**assembly capacity**' using imported cells.

US solar manufacturing capacity across the supply chain

GW



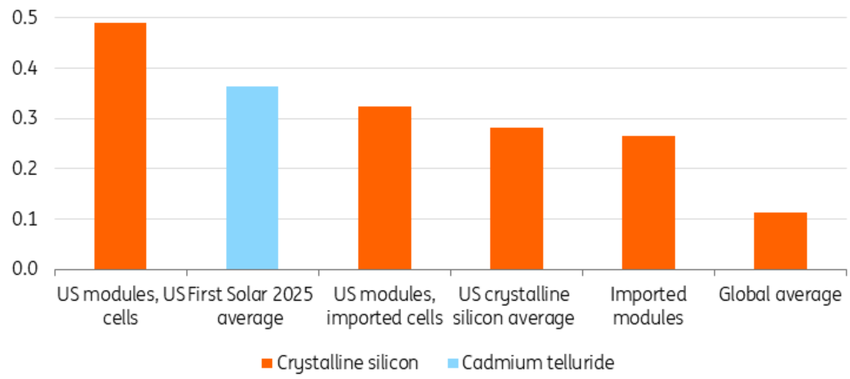
Source: ING Research, Solar Energy Industries Association

This creates additional risks. First, module manufacturers may need to produce way below nameplate capacity if they cannot secure inputs on time and at competitive prices. Second, reliance on imported cells exposes module manufacturers, and by extension solar developers, to tariffs and FEOC restrictions. By contrast, cadmium telluride modules, led by First Solar, benefit from about 14GW of fully integrated domestic capacity.

Another reason the US remains reliant on imports is price competitiveness. Even after accounting for tariffs, imported solar modules, a large share of which has ties to China, remain cheaper than US-made modules. It is also worth noting that even imported solar modules are over twice as expensive as the global average, where tariffs are minimal, suggesting the US manufacturing industry is far from competitive in the global market.

Solar module prices by technology and supply chain

cents/watt



Source: ING Research, Anza Renewables, First Solar, BNEF. Note: The average price of imported modules is inclusive of tariffs, but only as of the publication date of Anza's Q1 2026 market report.

These factors show that the US is far from fully localising its solar supply chain, though the 26GW of cell capacity under construction could improve the situation.

In the near term, however, many US projects will still rely on imports and be exposed to two policy risks: **tariffs** and **FEOC** requirements.

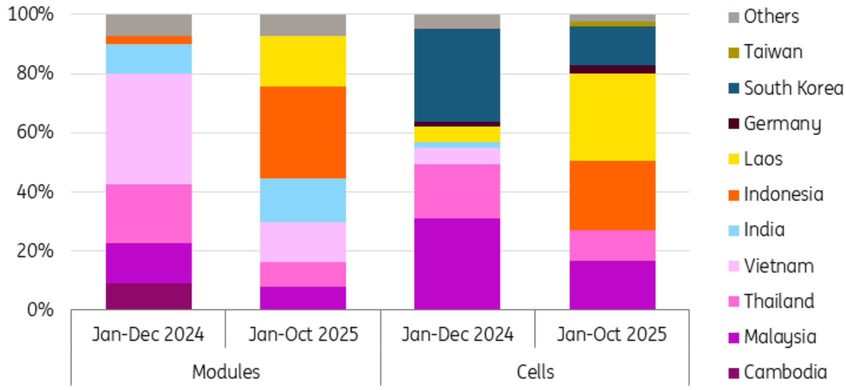
Tariffs continue to push China to find new trade reroutes

To curb tariff circumvention by China, the US sharply increased countervailing and anti dumping duties on solar imports from Cambodia, Thailand, Malaysia, and Vietnam in early 2025. This prompted China to react. At the time, we noted in a research article that alternative trade paths were emerging in Laos and Indonesia. Now, not only has 2025 trade data confirmed this move, but it has shown how fast rerouting happened.

From 2024 to 2025, the combined import share of modules from Laos and Indonesia jumped from 3% to 40%, with the cell import share from these two countries surging from 5% to 53%. In contrast, the combined import share from Cambodia, Malaysia, Thailand and Vietnam collapsed.

US imports of solar modules and cells

% (based on $)



Source: ING Research, BNEF

This rapid shift triggered a new US investigation. In February 2026, the Department of Commerce announced preliminary countervailing duties of 80.67%-143.3% on solar imports from Indonesia and Laos. It also imposed a 125% tariff on Indian imports, citing evidence of Chinese transshipment.

China's manufacturing overcapacity will incentivise it to continue finding new trade routes. But doing so could become harder, as China's remaining options in Southeast Asia are limited to markets such as the Philippines. And there is potentially Africa. Recent reports suggest solar manufacturing in Africa is expanding, although it is unclear how much is influenced by China.

Looking ahead, there are more trade issues we are watching:

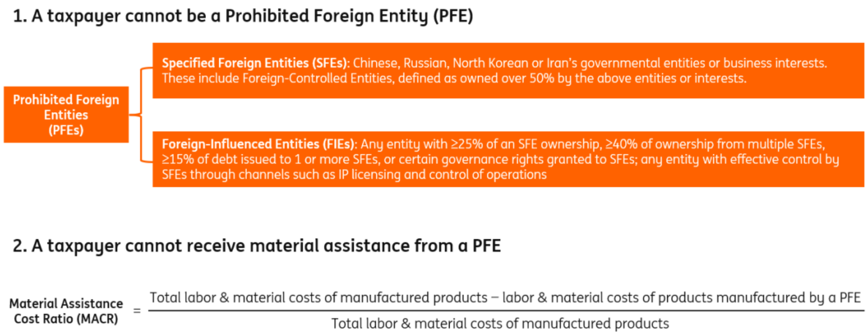
- **Court ruling on previous Southeast Asia tariff exemption:** In August 2025, the US Court of International Trade struck down a Biden-era rule that exempted for two years hefty tariffs on solar imports from Cambodia, Malaysia, Thailand, and Vietnam. The decision is now under appeal. If upheld, it could result in over $50bn in retroactive duties, dealing a major blow to US solar.
- **First Solar's patent lawsuit**: In March, the US International Trade Commission launched an investigation into whether several solar manufacturers – including Canadian Solar and Jinko Solar – are infringing First Solar's TOPCon (Tunnel Oxide Passivated Contact) patent. The case could take up to 18 months, but a ruling in First Solar's favour could significantly constrain solar equipment supply.
- **Appetite to shift toward non-China-linked suppliers**: There has been little increase in US imports from non-China-linked alternatives, such as South Korea, despite tariffs on Southeast Asian countries. It is worth seeing if that limited appetite can rise this year.

As developers map and rework their supply chains, they should focus not only on short-term fixes but also on durable solutions. For manufacturers, this means building partnerships up and down the supply chain to offer developers a more integrated model. That is the approach taken by Corning, Suniva, and Heliene, which together are establishing US made polysilicon, wafers, cells, and modules.

FEOC guidance prompts supply chain reevaluation – it's all about proving yourself

Another major policy risk stems from FEOC rules, which are designed to limit Chinese influence. FEOC compliance has two parts: developers cannot be prohibited foreign entities (PFEs), and they cannot rely on material assistance from such entities. Starting in 2026, companies with PFE ties will no longer qualify for clean energy tax credits.

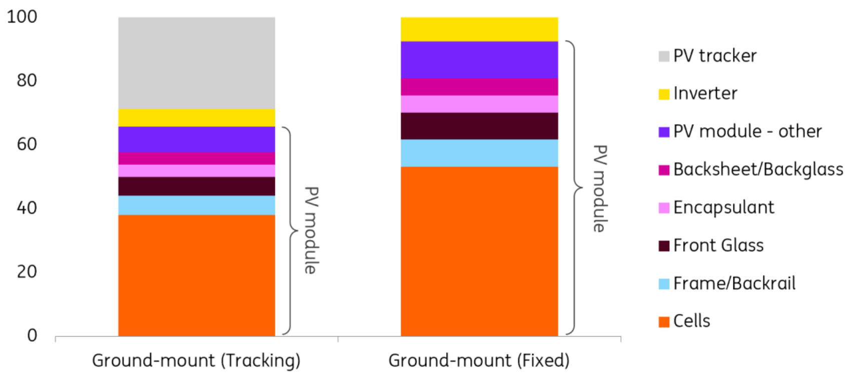
FEOC compliance criteria



Source: ING Research

In February, the Treasury Department issued draft guidance on material assistance. The key point is that solar and BESS developers only need to review a specific list of upstream components and ensure they are free of FEOC influence. The guidance adds tremendous certainty for developers to manage supply chains and advance projects.

Treasury's material assistance cost ratios to determine FEOC compliance



Source: ING Research, Treasury notices

However, in practice, ensuring FEOC compliance is not easy. Making sure PV module manufacturers are not PFEs could already be challenging. BloombergNEF estimates that over 20% of *US* solar module capacity may have high FEOC exposure through corporate structures or IP partnerships. That could be enough to fail the FEOC test.

Manufacturers that pass ownership and IP tests can still be disqualified if they receive material assistance from PFEs. Here, ensuring cells are compliant is the most critical step, as they account for the largest share of equipment costs used to assess FEOC status. But with limited compliant supply, this step can be challenging for some developers.

And full clarity will not come until the Treasury releases further guidance later this year to define PFEs more precisely and tighten rules against circumvention. A tougher stance means that Chinese firms might not be able to comply simply through restructuring ownership, financing, or contracts.

> Now, what developers need to do is not just verify ***whether*** they are FEOC compliant, but show robust proof of ***how*** they are compliant. To that end, supplier certifications do provide a certain level of comfort, as they are legally binding documents with penalties of perjury. But lenders are increasingly expecting more intense audits or third-party diligence. Detailed documentation is becoming critical to securing project financing.

Conclusion

Solar reshoring across the value chain is still progressing slowly, though momentum is building. Continued reliance on a China dominated global market, combined with tariffs and FEOC rules, means solar developers need significant work to make their supply chains bulletproof.

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Disclaimer

US battery storage needs innovation for long-term growth

US battery storage growth in 2026 looks strong, but supply chain risks could limit expansion. Rising demand for non Chinese components would push prices higher, while greater investment in technology and cost reduction is needed for the eventual subsidy phase-out

US BESS growth in 2026 looks strong. Still, efforts to avoid Chinese components could limit expansion, and technological innovation will be key to overcoming hurdles

Battery energy storage systems (BESS) are set to gain strong momentum in the US, as outlined in our Renewables Outlook for 2026. However, growth projections for 2026 vary widely, ranging from 48GWh (BloombergNEF) to 70GWh (Solar Energy Industries Association, or SEIA). Our view is closer to that of Wood Mackenzie, which expects installations to grow from 54GWh to 56GWh.

Where actual growth lands will depend largely on how well the industry manages supply chain risks – more specifically, shaking off its heavy reliance on China.

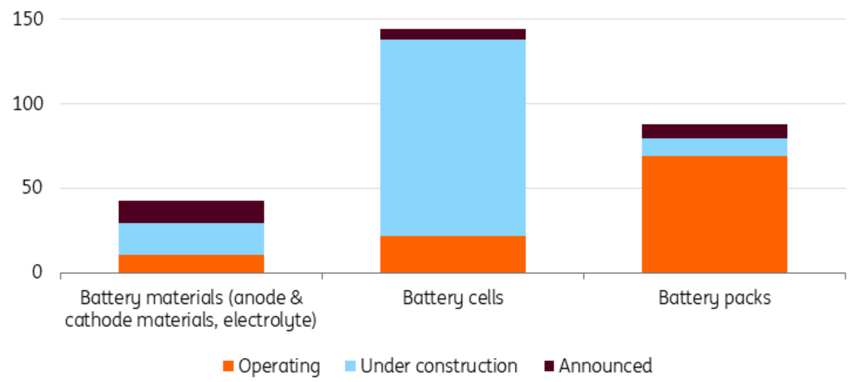
In the long term, however, structural cost reduction will be critical as tax credits phase out. We believe that is best achieved through technological innovation.

Supply chain gaps leave BESS projects vulnerable

Similar to solar, US BESS manufacturing is heavily concentrated in downstream assembly, with limited coverage of upstream components. SEIA estimates that while the US has 79GWh of nameplate battery pack manufacturing capacity as of early 2026, battery cell capacity stood at just 22GWh, highlighting a significant upstream gap.

US BESS manufacturing capacity across the supply chain

GWh



Source: ING Research, SEIA

Moreover, these figures include producers that make both stationary BESS and EV batteries, so the nameplate capacity dedicated to BESS alone is smaller. Plus, production can run well below nameplate capacity due to upstream bottlenecks. These two factors suggest that US BESS manufacturing has low self-sufficiency. Wood Mackenzie estimates that US battery cell manufacturing capacity could only meet 6% of domestic demand in 2025, though that share could increase to 40% by 2030. The scale of the projected ramp up is evident: 116GWh of battery cell capacity is currently under construction in the country.

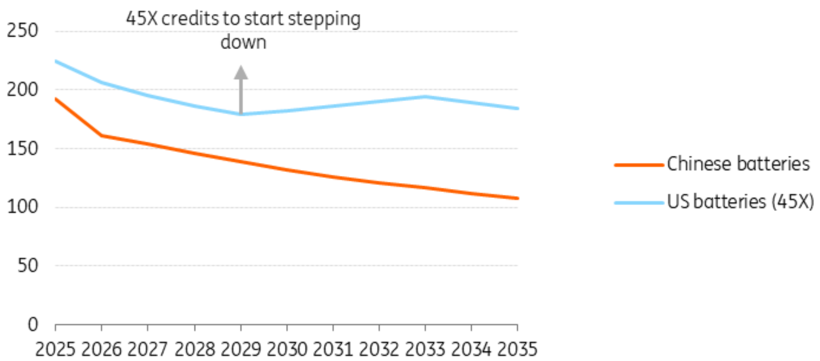
Tax credits are the lifeline for US BESS to trump cheap Chinese imports

Manufacturing and clean electricity tax credits are crucial to the growth of domestic BESS manufacturing. Without *either*, US producers would struggle to compete with low-cost Chinese imports.

Even under the current tariff regime, US four-hour turnkey battery systems using batteries that benefit from the 45X tax credits are still 31% more expensive than systems using imported Chinese batteries. The gap will further widen when the tax credits start to phase out.

Cost of US 4-hour turnkey battery storage systems

$/kWh, usable (real 2025)

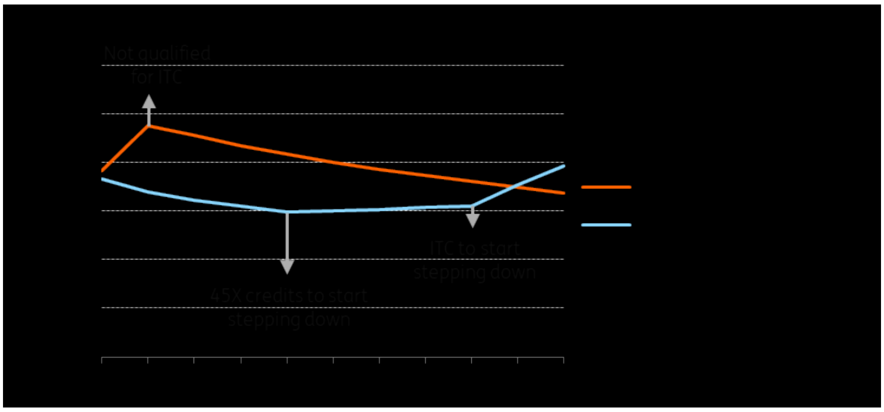


Source: BNEF, ING Research

It is only when the 45X tax credits are combined with 48E/48Y clean electricity tax credits in four-hour energy storage projects that US projects using US-made batteries can enjoy a cost advantage.

Cost of US 4-hour BESS projects

$/kWh, usable (real 2025)



Source: BNEF, ING Research. Note: Costs included in BESS projects but not turnkey storage systems include Engineering, Procurement, and Construction (EPC), grid connection, project overhead, and developer margin.

This means US battery manufacturers will need to double down on technological advancement and innovation. Based on current technology projections, tax credit phase-outs in the next decade will more than offset the cost advantage among US producers.

But before manufacturing capacity increases meaningfully across all supply chain components, the US BESS industry will remain exposed to tariffs and foreign entity of concern (FEOC) restrictions that target China. Below, we analyse these risks and how they can be managed.

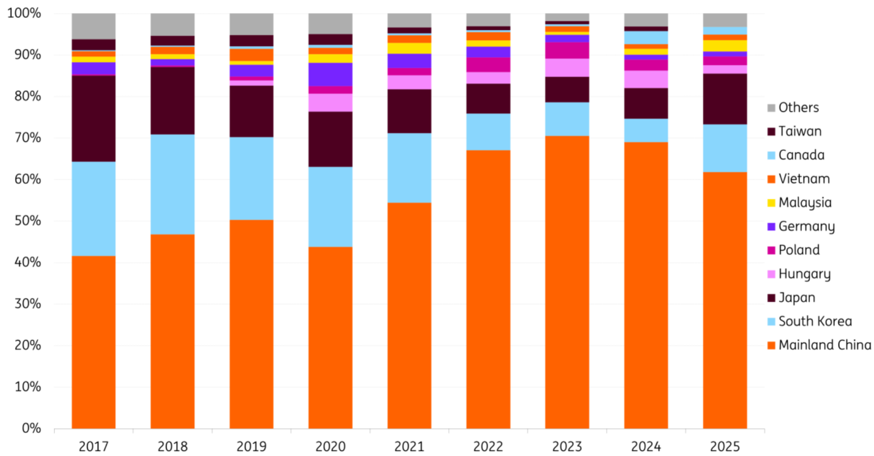
China is still king in US solar imports, but with a weakening role

It is difficult to quickly dismantle China's dominance, which accounts for nearly 90% of global lithium-ion battery manufacturing capacity. It will take time for the US to ramp up domestic manufacturing, and in turn, it will still need to rely on imports in the near term.

China accounted for the lion's share of US lithium-ion battery (both EVs and non-EVs) imports in 2025, at around 60%. Although import tariffs on Chinese batteries are higher than elsewhere, the difference has not been as material as seen in solar equipment. And because of safety control, batteries are harder to be transhipped via a third country. Rising tariffs have therefore led to a decreasing share of Chinese imports without much transshipment from proxy countries.

US import of lithium-ion batteries by origin

% (based on $)



Source: BNEF, ING Research

Here's what we'll be watching in terms of tariffs and trade looking ahead:

- **Further tariff hike on batteries**: It is worth watching whether tariffs on Chinese lithium-ion batteries, currently at 25% under Section 301 (to prevent unfair practices), will be raised further. This year, the US initiated an investigation under Section 301 after the Supreme Court struck down reciprocal tariffs imposed, citing the International Emergency Economic Powers Act (IEEPA).
- **Further tariff hike on graphite**: Graphite is a key metal in lithium-ion battery production. Natural graphite imports from China currently face a 25% tariff; any further hikes under Section 301 would mean higher supply chain costs for battery developers.

Because of limited battery transshipment from China and decent battery pack capacity in countries like South Korea and Japan, US BESS developers have some ability to shift away from Chinese imports. In fact, battery pack imports from South Korea and Japan rose in 2025 and could increase further this year.

The bigger constraint, however, is battery cells, where alternatives to China remain far more limited. We discuss this in more detail below.

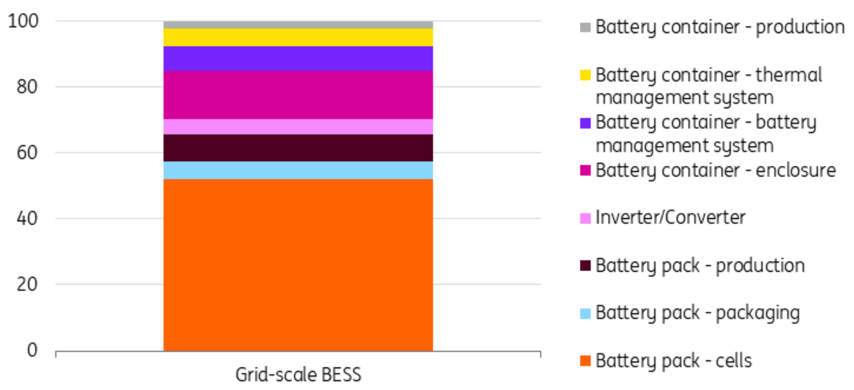
FEOC rules to have far-reaching impact on BESS supply chain

BESS developers will be subject to FEOC rules for longer, as BESS tax credits extend into the next decade, while solar and wind face near term eligibility deadlines. This means BESS developers will need to more structurally transition away from reliance on China.

In February, the Treasury Department released draft guidance clarifying how FEOC requirements can be met from a supply chain perspective. The Treasury is expected to issue further guidance on definitions of foreign prohibited entities (PFEs). Below is a summary of the BESS material assistance cost structure, while our detailed FEOC analysis can be found here.

Treasury's material assistance cost ratios to determine FEOC compliance

%



Source: ING Research, Treasury notices

If a developer now uses imported Chinese battery packs, it will need to replace them with US-made or non-Chinese imports. However, using US-produced battery packs does not eliminate FEOC risk. Developers must still verify that suppliers are free of Chinese influence across ownership, IP licensing, supply chain material assistance, etc.

Among these, battery cells are the most critical component, as they represent about half of the total cost used in the FEOC material assistance test for grid scale BESS.

This poses a significant challenge. China controls more than 80% of global battery cell production, while US cell capacity remains limited. On top of that, BloombergNEF estimates that roughly half of the current US operating BESS cell capacity may have high FEOC exposure risk due to corporate structure and IP partnership. That could further constrain near-term compliant supply.

Nevertheless, conditions can improve. Only 17% of planned US cell capacity is expected to face high FEOC exposure (excluding supply chain considerations), suggesting a meaningful expansion of compliant supply over time.

> In the near term, rising demand for FEOC compliant components is likely to push prices higher. Many developers may prioritise FEOC compliance over domestic content eligibility.

Longer term, progress will hinge on the pace of vertically integrated battery manufacturing growth in the US and among non Chinese partners.

Car manufacturers' repurposing EV batteries can boost BESS capacity

As EV tax credits are phased out under the One Big Beautiful Bill Act (OBBBA) and demand for battery storage from data centres accelerates, manufacturers are shifting their focus from EV batteries to BESS.

Ford, for example, has ended its EV battery joint venture with SK On and is repurposing its Kentucky plant to produce stationary LFP batteries, positioning the site as a BESS hub and exploring licensed CATL technology. GM is redirecting excess and used EV batteries to partners such as Redwood Materials and SunPower, supplying BESS for customers, including AI data centres. Stellantis's joint venture with Samsung SDI has made a similar shift in production priorities.

These strategic moves have reshaped industry expectations, prompting SEIA to double its forecast for US BESS capacity by 2030.

Conclusion

US BESS growth in 2026 looks strong, but efforts to avoid Chinese components could limit expansion. This underscores the need to scale domestic manufacturing across the value chain and, more importantly, to accelerate technology driven cost reductions to improve US competitiveness.

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LA PRODUCTIVITÉ RETROUVE DES COULEURS

Bruno Coquet et Éric Heyer

Résumé

La productivité du travail en France a traversé un long passage à vide entre la crise sanitaire et la fin 2022. Elle s'est ensuite redressée, progressant au rythme annuel de +1,3 % depuis le début de 2023 dans le secteur marchand non-agricole, soit plus rapidement que sa tendance de long terme (+0,9 % par an). ne comble cependant pas le retard accumulé au cours des trois années précédentes : la productivité par tête n'a pas encore tout à fait rejoint le niveau qui était le sien fin 2019, et elle demeure 5 % en dessous de sa trajectoire tendancielle.

Cette singularité française a fait l'objet de nombreuses conjectures. Celles-ci tenaient en partie aux données disponibles fin 2023 qui décrivaient une productivité déclinante mais qui révisées et complétées distinguent désormais clairement l'affaissement de la productivité en sortie de crise sanitaire du regain ultérieur.

Nous revenons sur cette séquence afin de mieux en comprendre les ressorts. La politique de soutien à l'apprentissage apparaît comme le principal déterminant de l'affaiblissement de la productivité, puisque qu'elle explique actuellement près de 30 % de l'écart à la tendance antérieure, suivie par la politique de soutien aux entreprises (17 %) et la baisse du coût du travail principalement liée au retard d'indexation des salaires aux prix (13 %) ; la baisse du chômage et ce qu'elle emporte en termes de profil des actifs (9 %) et la contraction persistante de la durée du travail (6 %) jouent également un rôle significatif. Au total nous parvenons à expliquer 75 % de l'écart de productivité accumulé depuis 2019 par rapport à la tendance de long terme.

L'usage est de mesurer la productivité du travail en attribuant l'ensemble de la valeur ajoutée marchande aux seuls salariés, comme si les non-salariés n'en avaient aucune. Mais la part de ces derniers s'accroît vigoureusement depuis 20 ans, et leurs caractéristiques se transforment (revenu, durée du travail, etc.). L'incorporation des non-salariés dans nos équations n'améliore cependant pas la qualité de celles-ci et modifie peu la hiérarchie des différentes variables identifiées dans notre modèle de base.

SciencesPo

La panne de productivité : une spécificité française

Au plus fort de la pandémie, la chute de la productivité du travail[1] a été plus prononcée en France que dans le reste de la zone euro. Le rebond a été vif dès le second semestre 2020, mais moins en France que chez nos partenaires européens (graphique 1).

À la fin 2024 la France n'a pas encore tout à fait rejoint le niveau de productivité par tête qui était le sien cinq ans plus tôt, alors qu'en moyenne l'ensemble formé par les autres pays de la zone euro est parvenu à ce résultat il y a quatre ans, dès le 1er trimestre 2021. Depuis deux ans notre pays comble progressivement le retard accumulé, en partie aidé par le ralentissement de nos voisins.

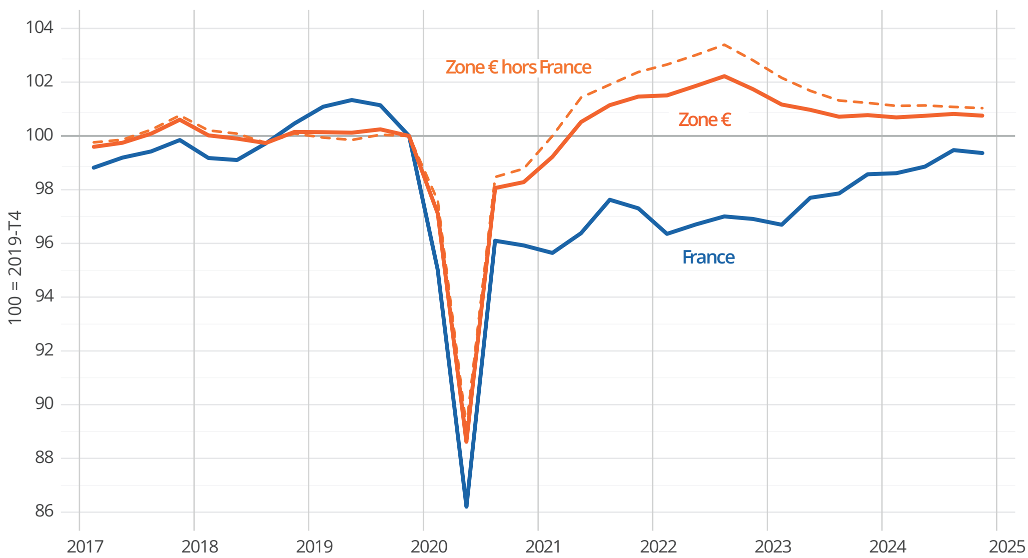
Entre fin 2019 et fin 2024 le contenu en emploi de la croissance a été très élevé en France : l'emploi salarié marchand a crû de +6,1 %, alors que la valeur ajoutée de ces secteurs ne progressait que de +5,5 %. Les performances en termes d'emplois des autres pays de la zone euro (+4,3 %) étaient, quant à elles, inférieures à celles de l'activité (+5,1 %).

Le très fort contraste qui prévaut entre 2020 et 2022 s'atténue ensuite (graphique 1), mais les mauvaises performances de l'Allemagne et de l'Italie pèsent alors fortement sur la moyenne de nos partenaires. En d'autres termes les emplois créés ou maintenus en France étaient moins productifs qu'auparavant, et moins productifs que ceux créés ou maintenus chez nos voisins. Il est également possible de lire l'exception française comme les deux faces des politiques publiques mises en œuvre au cours de cette sous période 2020-2022 : d'une part des instruments de soutien à l'offre plus prolifiques en termes d'embauches que de production et ayant bénéficié à des entreprises déjà fragiles[2], et d'autre part des mesures de soutien à la demande n'ayant pas produit les résultats escomptés, même si sur ce dernier point le comportement de « sur-épargne » des ménages n'est que légèrement plus prononcé que chez nos voisins.

1. Sauf mention contraire la productivité est celle des secteurs marchands non-agricoles.

2. Cazenave (2025).

Graphique 1. **Productivité par tête : France et zone euro (2017-2024)**

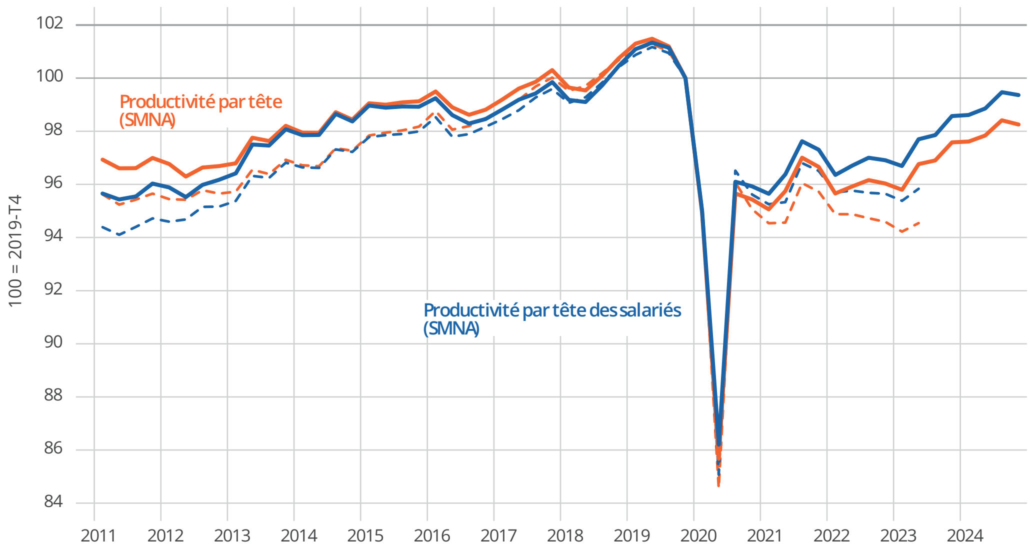


Calculs sur données Eurostat et Insee.

Note : secteurs marchands non agricoles.

Il reste que même si cette bulle d'emplois inexpliqués s'est légèrement dégonflée à mesure de la révision des comptes trimestriels et à la suite du rebasage des comptes nationaux en 2024, elle garde encore une part de mystère.

Graphique 2. **Productivité par tête, et révisions en France (2011-2024)**



Calculs sur données Insee (Comptes Nationaux trimestriels).

Note : secteurs marchands non agricoles.

Lecture : Les lignes en trait plein représentent la productivité telle qu'elle peut être calculée avec les données actuelles. Les lignes pointillées correspondent aux mêmes indicateurs de productivité que les lignes en trait plein, mais calculés avec les Comptes Nationaux trimestriels du deuxième trimestre 2023.

Après un choc et deux années au ralenti la productivité s'est redressée

La nouvelle base des comptes nationaux adoptée en 2024 et la succession des révisions trimestrielles habituelles ont sensiblement modifié le profil et le diagnostic que l'on peut faire des années écoulées depuis le début de la pandémie : d'une part le niveau de la valeur ajoutée marchande a été sensiblement réhaussé, d'autre part le niveau de l'emploi a été légèrement abaissé à partir de la fin 2021. Et si la révision de la valeur ajoutée est assez régulière, celle de l'emploi salarié est plus heurtée sur la période examinée ici.

Fin 2024 la productivité par tête restait encore légèrement inférieure (-0,6 %) à son niveau d'il y a cinq ans (graphique 2). L'impact de la crise sanitaire sur le niveau de la productivité a ravivé les débats concernant l'ampleur et la persistance de l'écart entre les comportements habituels d'emploi et ceux que l'on observe depuis 2008 : s'agissait-il ou non d'une nouvelle rupture marquant un ralentissement structurel – donc permanent – du rythme de productivité ? La productivité rejoindrait-elle sa tendance antérieure, compensant l'ensemble des pertes enregistrées durant la pandémie ? ou bien observerait-on le point bas de 2020 comme un choc permanent sur le niveau de productivité, celle-ci retrouvant ensuite sa tendance antérieure ?

Avant la pandémie, la croissance tendancielle de la productivité du travail était évaluée à +0,9 % par an. Les dernières ruptures de tendance de productivité remontent aux années 1990 et 2000, en lien avec les politiques d'allègement de cotisations et de réduction du temps de travail qui ont enrichi la croissance en emplois. La productivité par tête a ainsi fortement ralenti : sa croissance est passée de +2 % en rythme annuel dans les années 1980 à +1,4 % dans les années 1990, puis +0,9 % dans les années 2010[3].

Trois constats se dégagent des données désormais disponibles :

- Avant la crise sanitaire : la productivité par tête des salariés des SMNA évoluait à un rythme très ralenti depuis 2015 (+0,3 % par an en moyenne), suggérant une croissance déjà fortement enrichie en emplois (CICE, Pacte de responsabilité, etc.).
- Au sortir de la crise sanitaire : fin 2020 le niveau de productivité était de -4,1 % inférieur à celui atteint fin 2019.
- Après la crise sanitaire : la productivité par tête s'est accrue, d'abord lentement de +0,5 % par an en moyenne en 2021 et 2022, puis plus vivement et au-delà de sa tendance, à rythme annuel moyen de +1,3 % depuis le début 2023.

L'emploi reste élevé en regard du niveau de la production

Ce retour à une certaine normalité n'efface pas le passage à vide aussi prononcé que singulier de la productivité française entre les seconds semestres 2020 et 2022.

Entre fin 2019 et fin 2024 la faible croissance de la valeur ajoutée des SMNA (+1,1 % en moyenne par an) aurait dû aboutir, toutes choses égales par ailleurs, à une hausse modeste de l'ordre de +18 000 emplois salariés par an en moyenne dans ces secteurs (soit +89 000 en 5 ans) (tableau 1).

Le niveau de l'emploi salarié des SMNA s'est avéré très supérieur à ce qu'anticipaient nos modèles de prévision basés sur les déterminants habituels de la demande de travail[4] : en effet 1,071 million (+6,1 %) d'emplois salariés marchands ont été effectivement créés selon les chiffres de la comptabilité nationale[5], dont 982 000 s'expliquent soit par des comportements atypiques autour des déterminants habituels qui figurent dans nos équations (durée et coût du travail), soit par des facteurs exceptionnels, ou par une combinaison de ces différents facteurs.

3. Ducoudré & Heyer (2017). Précisons aussi que la baisse de la tendance de productivité (entre autres liée au ralentissement du progrès technique) n'est pas une spécificité française.

4. Ducoudré & Heyer (2017), Ducoudré (2019).

5. Cette étude a été réalisée avec les données disponibles fin février 2025.

Tableau 1. **Valeur ajoutée et emploi salarié SMNA**

Évolutions attendues et observées (2019-T4 à 2024-T4)

Secteurs Marchands Non-Agricoles / Salariés		Industrie	Construction	Services	Total
Tendance de la productivité par tête / an (%)	(a)	+1,6	-0,0	+0,9	+0,9
Valeur ajoutée (%)					
Évolution cumulée observée de la VA		-1,0	-7,3	+8,1	+5,5
Emploi (%)					
Évolution cumulée attendue	(b-5*a)	-8,7	-7,1	+3,6	+1,0
Évolution cumulée observée	(c)	+2,4	+5,5	+7,0	+6.1
Emplois cumulés à expliquer	(c-b-5*a)	+11,1	+12,6	+3,6	+5,4
Emploi (milliers)					
Évolution cumulée attendue	(d)	-249	–108	+447	+89
Évolution cumulée observée	(e)	+68	+84	+919	+1071
Emplois surnuméraires à expliquer	**(e-d)**	**+317**	**+193**	**+472**	**+982**

INSEE, Ducoudré & Heyer (2017), auteurs.

Lecture : les valeurs sont arrondies à la deuxième décimale, ce qui peut faire apparaître des écarts significatifs dans ce tableau simplifié, notamment entre les données annuelles et quinquennales en %.

Au sein de cette enveloppe, tous les secteurs ont contribué aux créations d'emplois quelle que soit le dynamisme de leur valeur ajoutée : +68 000 emplois ont été créés dans l'industrie là où nos modèles anticipaient -249 000 destructions d'emplois salariés en près de cinq ans, le secteur de la construction a créé +84 000 emplois alors que l'on pouvait s'attendre à ce que -108 000 postes disparaissent, et les services marchands ont créé +919 000 nouveaux emplois quand l'activité en aurait justifié seulement +447 000 (tableau 1).

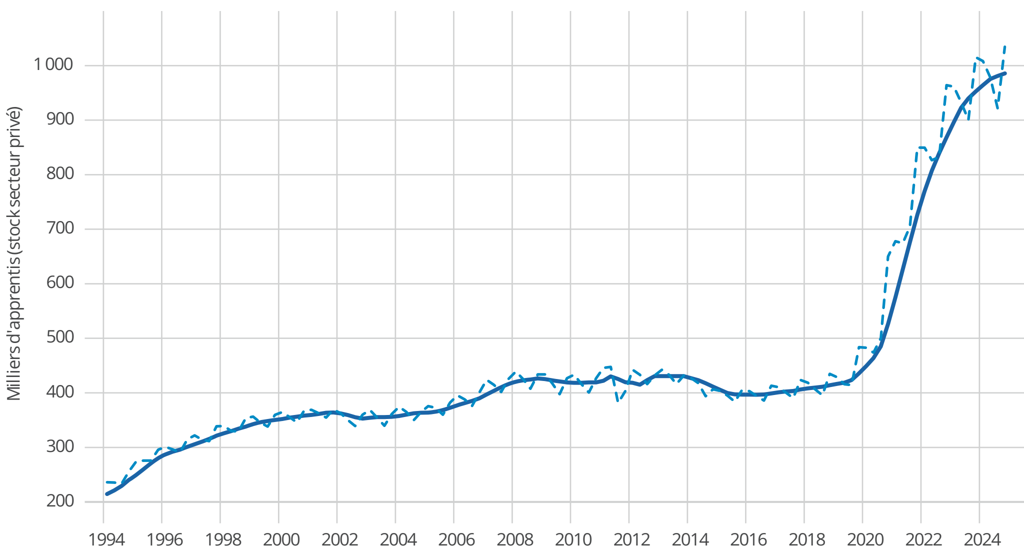
Les subventions à l'offre ont stimulé la demande de travail

L'écart cumulé depuis fin 2019 entre l'emploi observé et l'emploi attendu a culminé à 1,167 million au début 2023. Il s'est ensuite progressivement réduit, mais demeurait très élevé avec 982 000 emplois salariés surnuméraires fin 2024. Il est utile de comprendre pourquoi l'économie française a créé ou sauvegardé autant d'emplois: en effet, selon qu'ils sont le fruit de la mise en place de comportements nouveaux ou de stimuli ponctuels, que le coût en a été supporté par les finances publiques (via une dégradation du solde public), les ménages (via une baisse du salaire réel) ou les entreprises (via une baisse de leur taux de marge), les emplois créés durant cette période pourraient être plus ou moins durables.

Aujourd'hui, la nouvelle base des comptes nationaux ainsi que la disponibilité de données plus récentes sur les politiques publiques, permettent d'actualiser et de préciser nos travaux antérieurs.

En 2023 nous avions identifié que le contenu atypique de la croissance en emplois apparaissait pour l'essentiel consécutif aux mesures exceptionnelles prises afin de limiter l'effet négatif des confinements et de la crise sanitaire sur le marché du travail, dont certaines avaient été prolongées dans le but de soutenir l'élan de la relance économique une fois passé l'acmé de la crise. Trois facteurs expliquaient alors les deux tiers des emplois surnuméraires : la baisse de la durée du travail (horaires réduits, activité partielle, arrêts maladie, surcroît d'absentéisme, etc.), les politiques d'aide à l'apprentissage et de soutien aux entreprises (PGE, création d'entreprise, etc.).

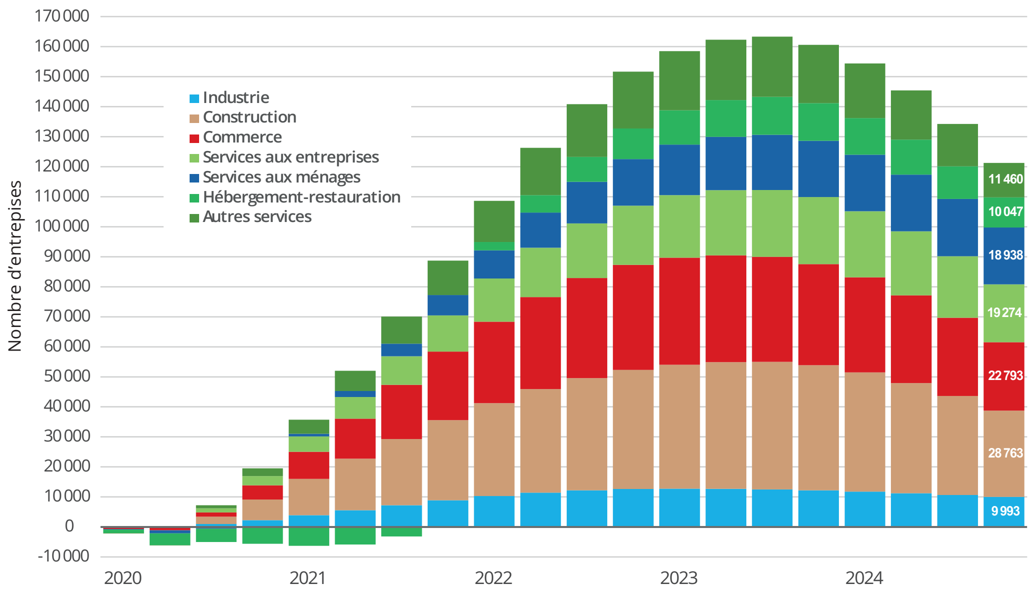
Graphique 3. **Apprentis en emploi (1994-2024)**



DARES, calculs des auteurs.

Note : Apprentis du secteur privé. Moyennes trimestrielles et mm4.

Graphique 4. **Évaluation du nombre d'entreprises « zombies » (2000-2024)**



Calculs sur données Insee, Dares, Banque de France, auteurs.

Lecture : le nombre d'entreprises « zombies » a culminé à 163 300 au troisième trimestre 2023, avant de reculer à 132 700 fin 2024 ; à cette date c'est le secteur de la construction qui comptait le plus d'entreprises de ce type (28 763).

Ces déterminants demeurent les principaux facteurs explicatifs du fort contenu en emploi de la croissance, mais les nouvelles données disponibles permettent de préciser nos précédentes estimations, et d'expliquer près de 75 % des créations d'emplois inattendues :

- L'apprentissage reste le principal déterminant de la dynamique de l'emploi salarié. La réforme structurelle qui a remis à plat cette politique en 2018 a facilité et stimulé le recours à ce dispositif[6]. La création en 2020 d'une « aide exceptionnelle » pour l'embauche d'apprentis, aide à la fois très généreuse[7] et non ciblée, dans le cadre du Plan de relance à mis en lumière les avantages et les nombreuses autres aides dont bénéficie ce dispositif, suscitant un engouement de toutes parts, apprentis, employeurs, organismes de formation. La légère réduction de l'aide exceptionnelle décidée début 2023 n'ayant que marginalement modifié l'équation économique des jeunes et des entreprises bénéficiaires, l'apprentissage a enchaîné les records jusqu'à parvenir à 879 000 nouveaux contrats conclus et 1,042 million d'apprentis en cours de formation en fin d'année en 2024, soit plus du double par rapport à fin 2019 (graphique 3). Au total nos estimations montrent que fin 2024 +284 000 emplois salariés doivent leur existence à cette politique de soutien à l'apprentissage (graphique 5), conjuguant l'aide à l'embauche mais aussi les autres financements publics (coûts pédagogiques, exonérations sociales et fiscales, etc.) qui représentent plus des trois quarts des aides dont bénéficient les contrats d'apprentissage au titre de la politique d'emploi. L'effet de ces financements que nous mesurons au travers du stock d'apprentis permet d'incorporer les facteurs qui soutiennent l'apprentissage sans instrumenter directement la réduction du coût du travail (exonérations et salaire réduit à une fraction du SMIC), ni le surcroît de demande qu'engendre le déficit public dû au financement des coûts pédagogiques[8]. Le nombre actuel d'apprentis dépend largement de la prolongation de ces aides[9].
- Les aides aux entreprises sont le deuxième facteur explicatif, dont l'importance s'est accrue jusqu'à la fin 2023 et qui reflue depuis. Ces dispositifs actés dans le

6. Coquet (2023, 2024).

7. Cette aide avait initialement été calibrée à 8 000€ en 2020 pour réduire à zéro le coût du travail de la plupart (85 %) des apprentis durant la première année de leur contrat. Elle a été abaissée à 6 000€ en 2023 sans que cela ne semble enrayer le rythme des entrées en apprentissage, suggérant que les autres aides soutiennent fortement l'attractivité du dispositif. Le montant total des ressources publiques consacrées à l'apprentissage avoisine désormais 24 Md€ par an ; pour la montée en charge depuis 2017 cf. Coquet (2023, 2024).

8. Le déficit cumulé de France Compétences pour les années 2021 à 2024 est estimé à 10,5 Md€ par la Cour des Comptes, qui auraient en théorie dû est financés par la taxe d'apprentissage.

9. Même en considérant que la réforme de l'apprentissage de 2018 a produit des effets bien plus importants que ceux anticipés par l'étude d'impact de la loi, il est possible de calculer qu'environ 210 000 emplois d'apprentis se seraient substitués à des formules contractuelles classiques, et ne sont donc pas comptabilisés dans les emplois surnuméraires.

sillage de la crise sanitaire afin de soutenir les créations d'entreprises ou de limiter les faillites ont pu enrichir la croissance en emplois dans des entreprises bien portantes, et aussi pu en aider d'autres à maintenir un niveau d'emploi supérieur à ce que justifiait leur activité (par exemple en attendant que cessent les ruptures d'approvisionnements ou pour se prémunir des difficultés de recrutement). La chute très prononcée des défaillances d'entreprises au cours des années 2020-2022 illustre quant à elle que des entreprises déjà fragiles et produisant peu ont survécu plus longtemps grâce au ballon d'oxygène des aides publiques, et ont de ce fait maintenu des emplois peu productifs. Sous l'effet conjugué de l'arrêt des aides, de la nécessité de rembourser le PGE[10] et de la conjoncture, le nombre d'entreprises « zombies » se contracte rapidement (graphique 4). Nous évaluons à +170 000 le nombre d'emplois surnuméraires liés à ces aides à la fin 2024 (graphique 5). De nombreux travaux ont cherché à évaluer l'impact des différentes mesures de soutien à l'activité (activité partielle, report de charges, prêts garantis par l'État, fonds de solidarité à destination des entreprises, bouclier tarifaire, etc.) sur les faillites d'entreprises[11], mais l'incidence de la diminution des défaillances sur l'emploi à la suite de la crise sanitaire est en revanche peu documentée alors que différer les faillites engendrerait une élévation plus forte du chômage à court terme en raison de destructions d'emplois finalement plus nombreuses[12]. Ces constats confortent l'idée de destructions d'emplois différées, et nos résultats qui montrent que ces aides ont significativement renforcé le contenu de la croissance en emplois.

- La baisse du coût du travail, principalement liée à une hausse moyenne des salaires inférieure à celle de l'inflation expliquerait quant à elle +129 000 emplois surnuméraires (graphique 5). La sensibilité de la demande de travail à son coût obtenue ici est proche de 0,3, voisine de celle estimée de manière agrégée figurant dans les modèles macro-économiques[13]. La formulation retenue ici incorpore par construction la baisse de coût du travail liée au salaire minimum des apprentis qui représente une fraction du SMIC selon leur âge et leur ancienneté dans le contrat, mais elle ne tient pas compte de l'ensemble formé par l'aide à l'embauche et les exonérations de cotisations sociales (salariales et patronales), ces effets étant capturés par la variable apprentissage. De même cette variable n'inclut pas des dispositifs qui tels l'Aide à l'Embauche des Jeunes[14] ont pu réduire transitoirement le coût du travail.
- La baisse du taux de chômage a également une influence significative. Cette variable joue le rôle d'un proxy agrégeant différents effets d'un marché du travail plus tendu, les difficultés de recrutement favorisant l'embauche de salariés relativement moins qualifiés, moins expérimentés, davantage exclus de l'emploi lorsque le chômage est élevé, et embauchés dans des emplois à plus faible valeur ajoutée. Le taux de chômage peut également interagir avec l'allégement du coût du travail des bas salaires qui favorise les modèles économiques à faible valeur ajoutée indépendamment de la qualification des nouveaux embauchés, comme pourrait le suggérer la concentration des emplois autour de 1,3 SMIC[15]. Au total la diminution du chômage a expliqué jusqu'à +140 000 emplois surnuméraires mi-2023, et encore +87 000 aujourd'hui (graphique 5), cet amenuisement reflétant la stabilisation du taux de chômage depuis le début 2022. Notre méthode d'estimation multifactorielle[16] permet probablement d'écarter que les effets associés à l'apprentissage (salaire égal à une fraction du SMIC, faible productivité de l'apprenti et productivité affaiblie du maître d'apprentissage, etc.) soient captés par cette variable taux de chômage, d'autant que si le boom de l'apprentissage a eu un effet sur l'activité et l'emploi des jeunes il n'a pas réduit le nombre de chômeurs dans cette tranche d'âge[17].

10. Au total 143 Md€ on été accordés au titre des PGE. Le PLF 2025 est construit sur la base d'un taux de défaut estimé à 4,11 %, soit une charge totale des appels en garantie de l'État de l'ordre de 5,9 Md€ dont 1,2 Md€ (20 % du total) à décaisser en 2025 et après.

11. Kalemli-Özcan *et al.* (2020), Guérini *et al.* (2020), OCDE (2020) ; Schivardi et Romano (2021) ; Banerjee *et al.* (2020a).

12. Banerjee *et al.* (2020b) Heyer (2020).

13. Cette sensibilité de la demande de travail à son coût est très proche de celle évaluée dans d'autres travaux économétriques (pour des références voir Heyer (2023).

14. Actée dans le Plan de Relance de 2020, la dépense s'est montée à 900 millions d'euros entre 2020-S2 et 2022-T1.

15. Bozio et Wasmer (2024).

16. C'est une différence d'approche avec la méthode retenue par la Devulder *et al.* (2024) qui additionne les effets des différents déterminants estimés indépendamment les uns des autres.

17. Dans la tranche d'âge des 15-24 ans le nombre de chômeurs au sens du BIT a baissé de -1,1 % (-7 000) contre -10 % (-190 000) dans les tranches d'âge supérieures.

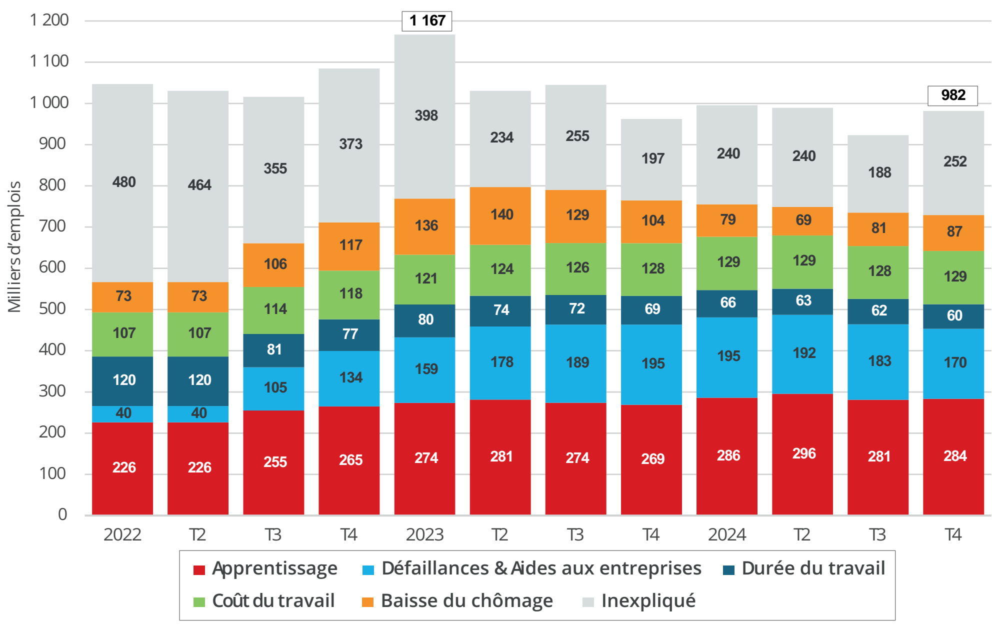
■ Enfin, les dispositifs de chômage partiel mis en place en France comme dans la plupart des pays européens ont protégé les salariés en faisant porter en grande partie l'ajustement sur les heures travaillées plutôt que sur le niveau d'emploi. Mais ces dispositifs exceptionnels ayant expiré au premier semestre 2022, ils n'expliquent pas totalement la diminution de la durée du travail qui a mis près de cinq ans à retrouver son niveau de fin 2019[18] dans la majorité des secteurs. Estimée par un coefficient commun, l'élasticité de la durée du travail à l'emploi est significative et sa valeur, autour de -0,5, proche des estimations obtenues par ailleurs[19]. La faiblesse de la durée du travail expliquait ainsi encore +120 000 emplois surnuméraires au début 2022, mais l'influence initialement importante de ce déterminant s'amenuise à mesure que la durée du travail se rapproche de son niveau d'avant pandémie. Malgré tout ce faible volume horaire de travail explique encore +60 000 emplois surnuméraires fin 2024 (graphique 5).

18. Cette plus faible durée du travail pourrait trouver une explication dans la hausse de l'absentéisme depuis la crise sanitaire et qui reste aujourd'hui toujours à des niveaux supérieurs à ceux observés en 2019.

19. Voir Heyer (2023) pour des références précises.

Ces variables laissent sans explication apparente 25 % des emplois surnuméraires créés ou maintenus depuis cinq ans. On peut néanmoins penser que la rétention d'emplois (emplois maintenus) ne s'explique pas seulement par les politiques publiques de soutien aux entreprises, mais aussi par des comportements atypiques de celles-ci : certaines auraient ainsi pu choisir de conserver des salariés en rognant sur leurs marges pour des raisons conjoncturelles (carnets de commandes remplis mais difficultés d'approvisionnement perçues comme transitoires, emplois très qualifiés difficiles à recruter à nouveau en cas de rebond dans un contexte de pénurie de main d'œuvre, etc.). La diminution des incertitudes, qui désormais écarte la probabilité d'un vif rebond économique à brève échéance devrait contribuer à un retour aux comportements habituels, d'autant que la situation financière des entreprises ne s'améliorerait pas, signifiant une réduction de cette rétention de main d'œuvre.

Graphique 5. **Déterminants des emplois surnuméraires (2022-2024)**



Insee, Dares, Banque de France, calculs des auteurs.

Note : secteurs marchands non-agricoles. Le nombre d'emplois à expliquer et la tendance de productivité contrefactuelle sont issus du tableau 1.

Lecture : le modèle 1 (encadré) décompose les déterminants expliquant 982 000 d'emplois surnuméraires en regard du nombre qui auraient été créés si le trend de productivité s'était prolongé sans choc sur son rythme de croissance antérieur à la pandémie. Fin 2024 la politique d'apprentissage expliquait 284 000 de ces emplois surnuméraires, contre seulement 269 000 fin 2023.

On peut aussi observer que les facteurs qui ont contribué à enrichir la croissance en emplois se contractent telle la durée du travail, mais aussi le soutien aux entreprises dont le volume diminue à mesure qu'elles remboursement leurs PGE ou que les faillites d'entreprises « zombies » se font plus nombreuses (graphique 4). L'apprentissage reste pour le moment à un niveau élevé, mais une normalisation de ce dispositif semble inévitable en raison des contraintes budgétaires, avec à la clé des destructions d'emplois d'apprentis et possiblement dans le secteur de la formation professionnelle. De même la hausse prévue du taux de chômage et le rattrapage en cours des salaires réels devraient progressivement réduire l'effet positif de ces variables sur l'emploi.

Décomposition sectorielle

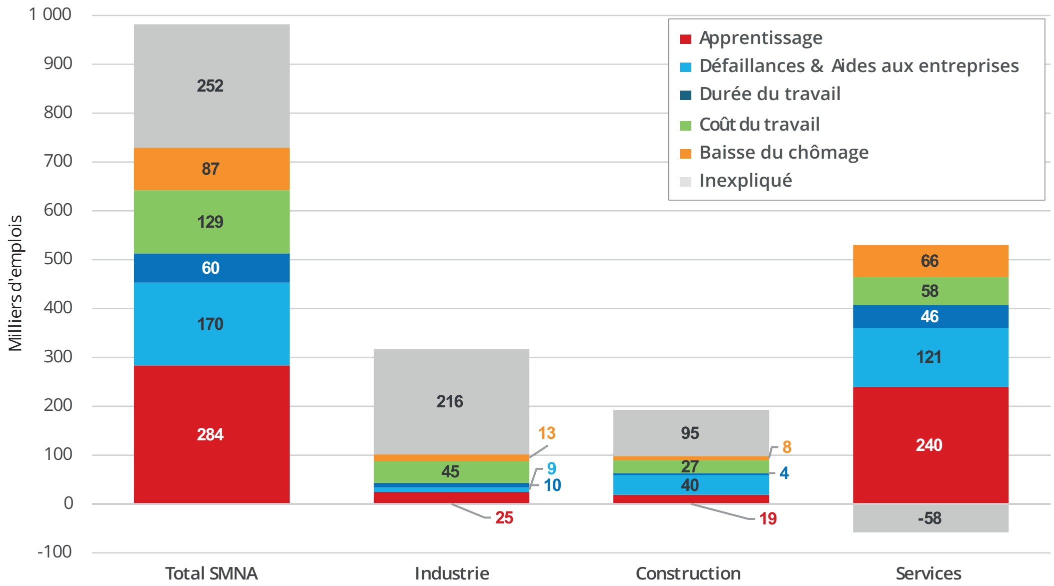
La méthode d'estimation que nous utilisons permet de décomposer les contributions des déterminants du surplus d'emploi par secteur d'activité (graphique 6). Les résultats sectoriels sont inégaux :

- C'est dans le secteur tertiaire que le caractère explicatif de notre modèle est le meilleur. On note que dans ces secteurs les créations d'emplois d'apprentis représentent plus de 40 % des emplois surnuméraires à la fin 2024. Les déterminants usuels et ceux identifiés au niveau agrégé du modèle 1 (encadré) expliquent la totalité des créations d'emplois surnuméraires par rapport à la tendance de productivité des services marchands.
- À l'opposé notre modèle explique seulement 32 % des créations d'emplois dans l'industrie : le coût du travail apparaît comme le principal déterminant des emplois surnuméraires comptabilisés dans ces secteurs, et l'apprentissage ne vient qu'en deuxième position (le nombre d'apprentis n'a progressé « que » de +67 % dans l'industrie entre 2019 et 2024, contre +183 % dans les services). Ce résultat pourrait être rapproché des informations selon lesquelles la rétention d'emplois à l'initiative et à la charge des entreprises a été particulièrement forte dans l'industrie, les entreprises hésitant à se séparer de leurs salariés qualifiés dans une période où coexistaient des difficultés d'approvisionnement et de recrutement (cf. ci-dessus).
- Dans le secteur de la construction ce sont les aides aux entreprises qui apparaissent comme le principal facteur explicatif du nombre inhabituel d'emplois créés[20], suivies par la baisse du coût du travail et de l'apprentissage. Notre modèle n'explique cependant que 51 % des emplois surnuméraires créés dans ce secteur[21].

20. Sachant que nous n'identifions pas directement les politiques de soutien de la demande (par exemple la rénovation énergétique) comme ayant pu stimuler l'emploi dans ce secteur.

21. Soulignons cependant que dans les secteurs de la construction les évolutions de la productivité sont de tout temps difficiles à expliquer. Un argument souvent avancé étant celui du travail non déclaré, difficile à mesurer et à relier à la valeur ajoutée dont on ignore si elle est dans ce cas déclarée ou non.

Graphique 6. **Déterminants des emplois salariés surnuméraires par secteur (2019-T4 à 2024-T4)**



Insee, Dares, Banque de France, calculs des auteurs.

Note : secteurs marchands non-agricoles. Le nombre d'emplois à expliquer dans chaque secteur, et la tendance de productivité contrefactuelle dans chaque secteur également sont issus du tableau 1.

Lecture : le modèle 1 (encadré) décompose les déterminants expliquant 982 000 d'emplois surnuméraires en regard du nombre qui auraient été créés dans chaque secteur d'activité si le trend de productivité du secteur s'était prolongé sans choc sur son rythme de croissance antérieur à la pandémie. L'apprentissage contribue à expliquer 240 000 emplois dans les services et 25 000 emplois dans l'industrie. Dans les services le pouvoir explicatif du modèle 1 est très fort, aussi le résidu peut être négatif (comme ici -58000 au 4e trimestre 2024) ou positif sans que cela ait une signification particulière.

Encadré 1. **Le modèle de demande de travail**

Modèle 1

L'équation de demande de travail des entreprises, écrite sous la forme d'un Modèle à correction d'erreurs (MCE), est issue de l'optimisation sous contrainte budgétaire (ici minimisation des coûts de production) d'une fonction de production à élasticité de substitution constante (CES).

L'emploi de la branche *i* dépend de la valeur ajoutée de la branche, du coût salarial réel de branche et d'une tendance de productivité du travail de l'emploi salarié total (*trend*).

Dans cette formulation, le progrès technique est neutre au sens de Harrod. Cela suppose qu'il ne porte que sur le facteur travail et assure un ratio capital sur production constant à long terme, condition nécessaire à l'existence d'un sentier de croissance équilibrée.

À long terme, compte tenu de la présence d'un *trend* de productivité, l'emploi est indexé unitairement sur l'activité mais l'existence de rigidités induit des délais d'ajustement à court terme.

Nous enrichissons cette formulation traditionnelle de l'équation de demande de travail, proche de celle estimée notamment dans Ducoudré, Heyer et Plane (2017) ou Heyer (2020), en y intégrant des données sur faillites d'entreprises ainsi que sur le nombre d'apprentis par secteur.

Du fait de leur évolution favorable malgré la forte baisse d'activité constatée, les défaillances d'entreprises sont ici un proxy des aides accordées aux entreprises en marge de la crise sanitaire. Les défaillances étant fortement corrélées à l'activité dans le secteur, nous avons dans un premier temps calculé un taux de défaillances, ratio du nombre de faillites à la valeur ajoutée du secteur[(a)], puis nous avons extrait sa tendance à l'aide d'un filtre Hodrick-Prescott que nous avons stabilisé à sa valeur de fin 2019 pour les années 2020-2024[(b)]. Afin de prendre en compte des effets dans leur intégralité, nous avons ensuite calculé l'écart cumulé entre le taux et sa tendance pour les défaillances d'entreprises pour chaque secteur.

(a) Une pondération par la taille des entreprises aurait été utile, mais les statistiques de défaillances d'entreprises de la Banque de France ne croisent pas taille et secteur. Il en est de même pour les créations d'entreprises, ce qui empêche de prendre en compte les microentreprises dans notre variable.

(b) Une alternative aurait pu consister à suivre la tendance observée. Notre hypothèse amplifie le phénomène : elle change le résultat quantitatif (écart absolu) mais pas le résultat qualitatif (sens de variation significatif) et augmente l'élasticité estimée et peu l'effet global.

Par ailleurs, l'estimation incorpore des effets fixes sectoriels et temporels. L'introduction de ce dernier dans l'équation permet de capturer l'effet agrégé d'un choc macroéconomique sur l'emploi. Cette spécification contrôle ainsi l'effet direct lié à une crise macroéconomique, à l'instar de la crise de la Covid-19, sur l'emploi. Les différentes élasticités estimées s'interprètent en complément de cet effet macroéconomique. Cela nous permet notamment de tester l'idée selon laquelle le marché du travail serait davantage impacté par une crise qui s'accompagne de faillites d'entreprises (Banerjee *et al.*, 2020b).

Nous avons estimé cette équation de demande de travail en panel sur la base des comptes nationaux de branches dans la nomenclature A17. Pour des raisons de disponibilité de données concernant les défaillances d'entreprises ainsi que les contrats d'apprentissage, sont exclus de l'analyse les secteurs de l'Agriculture (AZ) et de l'Administration publique, l'Enseignement, la Santé humaine et action sociale (OQ). Pour cette même raison, l'industrie n'est par ailleurs pas décomposée en sous-secteurs.

Au total, notre estimation comporte un panel de dix branches et a été réalisée sur la période 1995 2024, soit 1200 observations, à l'aide de la Méthode SUR (Seemingly Unrelated Regressions).

Dans ce MCE, une contrainte d'égalité sur les coefficients de long terme est imposée en fonction du regroupement des branches en nomenclature A17. Le coefficient de la force de rappel est également contraint de façon à s'assurer que ce dernier est bien négatif pour toutes les branches.

Modèle 2

Dans ce modèle, nous imposons le ratio entre non-salariés et salariés dans l'équation de long terme, et nous estimons l'effet de sa variation dans la dynamique de court terme.

Modèle 3

Ce modèle réplique le modèle 1 en remplaçant l'emploi salarié marchand par le total de l'emploi salarié et non salarié de ce même secteur.

Modèle 1 : modèle à correction d'erreur de la demande de travail salarié marchand

$$\Delta\log(Es_{i,t}) = \alpha - \gamma_{Es}.\left(\log(Es_{i,t-1}) - \log(VA_{i,t-1}) - \sigma_{Es}.\log\left(\frac{W_{i,t-1}.(1+tx_{cot.pat_{i,t-1}})}{Pva_{i,t-1}}\right) - \beta_{Es,i}.(\sigma_{Es}-1).trend - \delta_{Es}.\log(HLs_{i,t-1})\right)$$
$$+\ \kappa_{Es,i}.\sum_{j=1}^{n}\Delta\log(Es_{i,t-J}) +\ \tau_{Es,i}.\sum_{j=0}^{n}\Delta\log(VA_{i,t-j}) + \upsilon_{Es,i}.\sum_{j=0}^{n}\Delta\log\left(\frac{W_{i,t-j}.\left(1+tx_{cot.pat_{i,t-j}}\right)}{Pva_{i,t-j}}\right) + \varphi_{Es,i}.\sum_{j=0}^{n}\Delta\log(HLs_{i,t-j})$$
$$+\ \pi_{Es,i}.\sum_{j=0}^{n}EC_{TxDef_{ent_{i,t-j}}} +\ \omega_{Es,i}.\sum_{j=0}^{n}\Delta\log(Apprent_{i,t-j}) +\ \partial_{Es,i}.\sum_{j=0}^{n}\Delta U_{t-j} + \mu_i + \mu_t + \varepsilon_{i,t}$$

Modèle 2 : modèle à correction d'erreur de la demande de travail salarié et non-salarié marchand

$$\Delta\log(Es_{i,t}) = \alpha - \gamma_{Es}.\left(\log(Es_{i,t-1}) - \log(VA_{i,t-1}) + \log\left(1+\frac{Ens_{i,t-1}}{Es_{i,t-1}}\right) - \sigma_{Es}.\log\left(\frac{W_{i,t-1}.(1+tx_{cot.pat_{i,t-1}})}{Pva_{i,t-1}}\right) - \beta_{Es,i}.(\sigma_{Es}-1).trend - \delta_{Es}.\log(HLs_{i,t-1})\right)$$
$$+\ \kappa_{Es,i}.\sum_{j=1}^{n}\Delta\log(Es_{i,t-J}) +\ \tau_{Es,i}.\sum_{j=0}^{n}\Delta\log(VA_{i,t-j}) + \upsilon_{Es,i}.\sum_{j=0}^{n}\Delta\log\left(\frac{W_{i,t-j}.\left(1+tx_{cot.pat_{i,t-j}}\right)}{Pva_{i,t-j}}\right) + \varphi_{Es,i}.\sum_{j=0}^{n}\Delta\log(HLs_{i,t-j})$$
$$+\ \pi_{Es,i}.\sum_{j=0}^{n}EC\_TxDef_{ent_{i,t-j}} +\ \omega_{Es,i}.\sum_{j=0}^{n}\Delta\log(Apprent_{i,t-j}) +\ \partial_{Es,i}.\sum_{j=0}^{n}\Delta U_{t-j} + \psi_{Es,i}.\sum_{j=0}^{n}\Delta\frac{Ens_{i,t-1}}{Es_{i,t-1}} + \mu_i + \mu_t + \varepsilon_{i,t}$$

Données utilisées et mnémoniques

W : Salaire moyen brut par tête, en euros, INSEE
Pva : Prix de la valeur ajoutée, INSEE
Es : Emploi salarié, personnes physiques, INSEE
Ens : Emploi non-salarié, personnes physiques, INSEE
π : Productivité du travail par tête, INSEE
HLs : Volume horaire de travail des salariés, en heures, INSEE
VAvol : Valeur ajoutée en volume, INSEE
Txcotpat : Taux de cotisations patronales, INSEE (effectif)
$\boldsymbol{TxDef_{ent}} = \boldsymbol{Def_{ent}} / \boldsymbol{VA}$: Taux de défaillances d'entreprises ($\boldsymbol{Def_{ent}}$ = nombre de défaillances, Banque de France)
$\boldsymbol{EC\_TxDef_{ent}}$: Écart cumulé du taux de défaillances d'entreprises (par rapport à sa tendance mesurée par un filtre de HP)
U : Taux de chômage au sens du BIT, INSEE
Apprent : Stock d'apprentis, DARES, auteurs.

L’emploi non-salarié a eu peu d’effet dans la période récente

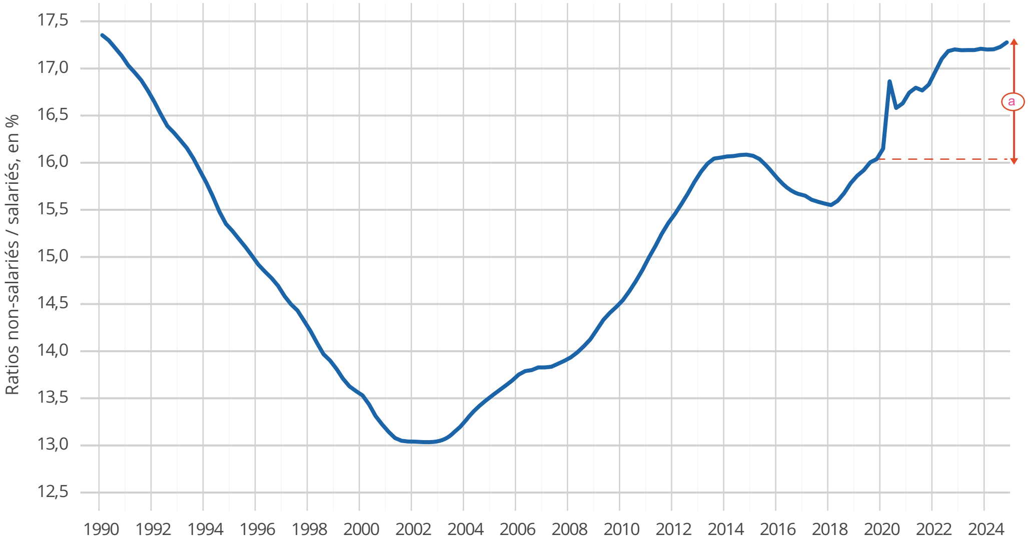
La croissance de la productivité du travail est usuellement identifiée à celle des salariés des SMNA, ce qui implique que l’ensemble de la valeur ajoutée marchande est attribuée à ces derniers, comme si les non-salariés n’en avaient aucune. Cette approximation est correcte tant que les ratios entre d’une part le nombre salariés et non-salariés, et d’autre part la production horaire de ces deux catégories d’actifs, sont constants.

Mais la décrue tendancielle du non-salariat s’est retournée au début des années 2000 (graphique 7), puis cette reprise s’est accélérée avec la création du régime des microentrepreneurs en 2008. Cette croissance du nombre de non-salariés a marqué le pas de 2015 à 2018 avant de reprendre à un rythme soutenu, les non-salariés représentant 27 % des créations d’emplois depuis 2019, plus de deux fois leur poids dans l’emploi total.

Cet enrichissement de la croissance en emplois non-salariés induit une productivité tendanciellement plus faible (+0,8 % par an, tableau 2) que celle mesurée avec les seuls salariés (+0,9 % par an, tableau 1). En effet les nouveaux non-salariés se recrutant plus fréquemment à la marge des catégories actives, leur profil est assez différent des indépendants « classiques » : leur revenu et leur durée du travail sont plus faibles, si bien qu’en moyenne la valeur ajoutée et la durée du travail moyennes de l’ensemble des non-salariés diminuent tendanciellement.

Enfin, même si la hausse du nombre de non-salariés est très forte, la nouvelle base des comptes nationaux a sensiblement modifié le partage entre créations d’emplois salariés (révisées à la hausse) et non-salariés (révisées à la baisse) induisant un effet sur la productivité de ces deux groupes d’actifs.

Graphique 7. **Ratio non-salariés / salariés (1990-2024)**



INSEE, calculs des auteurs. Secteurs marchands non-agricoles, emplois ETP.

Lecture : fin 2024 les emplois non-salariés (3,0 millions) représentaient 17,3 % de l’emploi salarié (17,4 millions), en hausse de 1,3 % par rapport à fin 2019. Si l’emploi salarié avait cru à la même vitesse que l’emploi non-salarié, ce ratio serait demeuré constant à 16 %, et on aurait comptabilisé 222 000 emplois (a) salariés de plus fin 2024.

Pour évaluer l'impact de cette transformation structurelle sur l'évolution récente de la productivité, nous faisons évoluer notre équation de demande de travail en adoptant deux stratégies différentes.

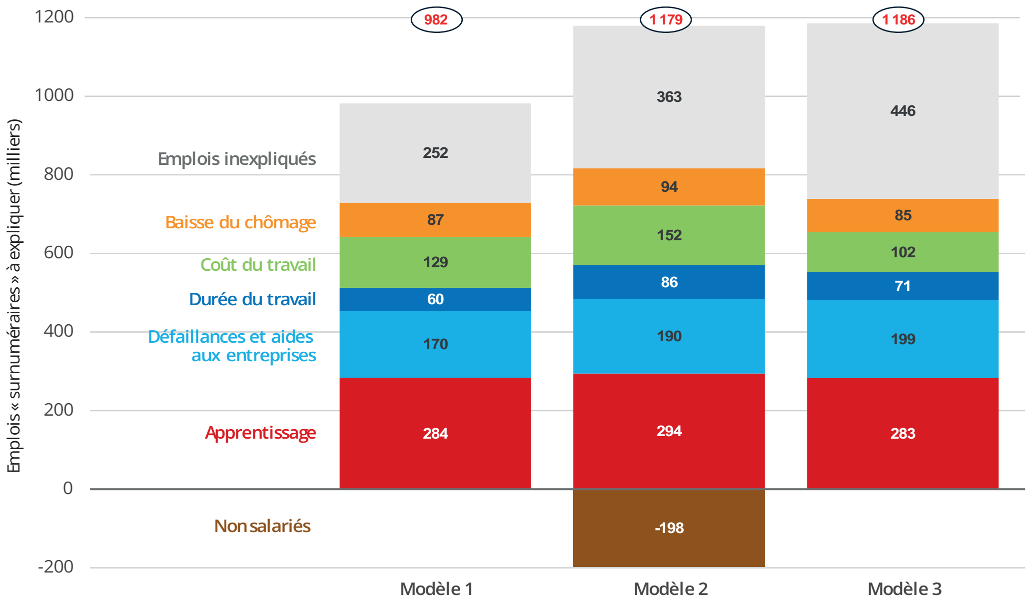
Le modèle 2 (encadré) estime la demande de travail salarié marchand comme le faisait le modèle 1, mais en ajoutant une variable illustrant l'influence de la hausse du taux de non-salariés sur l'excès d'emplois salariés marchands. Après une période de stabilité ce ratio a de nouveau enregistré une forte hausse depuis 2020. En faisant l'hypothèse qu'il se serait stabilisé à sa valeur de fin 2019, le surcroît de créations d'emplois salariés marchands qu'il aurait fallu expliquer par d'autres déterminants que ceux de notre équation standard de demande de travail aurait atteint 1,179 million. Le modèle estime en effet que 198 000 nouveaux non-salariés se sont substitués à des créations d'emplois salariés marchands, soit une valeur proche de la hausse du rapport entre non-salariés et salariés entre la fin 2019 et la fin 2024 (graphique 7). Si globalement l'ordre d'importance des variables explicatives n'est pas modifié par rapport au modèle 1, l'ampleur de leur influence est sensiblement amoindrie, si bien que le modèle 2 n'explique que 69 % du surplus total d'emplois salariés qui auraient été créés en l'absence d'une hausse de la part des non-salariés dans l'emploi. Comme en outre cette spécification considère comme exogène les créations d'emplois non-salariés, sans expliquer les raisons de leur dynamisme (coût relatif du travail, aides aux créateurs d'entreprises, etc.), elle apparaît à tous points de vue moins satisfaisante que le modèle 1 tout en étant plus complexe.

Le modèle 3 est identique au modèle 1 mais identifie les déterminants des créations d'emplois marchands surnuméraires sans distinction de statut. Nous cherchons à expliquer l'emploi total (salariés + non-salariés) des SMNA, soit un surnombre de 1,186 million d'emplois par rapport à ce que laissait attendre la croissance de la valeur ajoutée marchande : la valeur ajoutée est la même que dans le modèle 1, mais la variable d'emploi est ici l'emploi total des SMNA (salariés + non-salariés), et la tendance de productivité est plus faible (tableau 2). A nouveau l'ordre d'importance des différentes variables explicatives n'est pas profondément modifié, si ce n'est l'influence plus réduite de la durée du travail. Le modèle n'explique cependant que 62 % des créations d'emplois surnuméraires, résultat sensiblement inférieur à celui des deux autres modèles (graphique 8).

En première approche la manière dont nous avons pris en compte le dynamisme des créations d'emplois non-salariés dans la période récente ne nous permet pas de mieux expliquer le contenu en emploi de la croissance.

L'influence des emplois non-salariés pourrait faire l'objet d'une investigation spécifique, notamment si l'on retient l'idée qu'ils se substituent en partie à des emplois salariés, possiblement en raison d'un coût relatif beaucoup plus avantageux pour les employeurs, qui pourrait inciter ces derniers à sous-traiter du travail à des micro-entrepreneurs plutôt que d'embaucher des salariés en contrat court ou des intérimaires dont le coût du travail est plus élevé.

Graphique 8. **Contributions aux créations d'emploi surnuméraires incluant les non-salariés**



Insee, Dares, Banque de France, calculs des auteurs.

Note : secteurs marchands non-agricoles.

Les nombres d'emplois à expliquer et les tendances de productivité contrefactuelles retenus pour chaque modèle sont issus des tableaux 1 et 2. Le nombre d'emplois à expliquer par rapport à la tendance de productivité d'avant-pandémie est le nombre de salariés pour les modèles 1 et 2, et l'ensemble salariés et non-salariés pour le modèle 3.

Lecture : le modèle 1 décompose les déterminants expliquant 1,179 million d'emplois salariés surnuméraires en regard du nombre qui auraient été créés si le trend de productivité s'était prolongé sans choc sur son rythme de croissance antérieur à la pandémie. L'apprentissage contribue à expliquer 284 000 emplois, les aides aux entreprises 170 000.

Tableau 2. **Valeur ajoutée et emploi total SMNA**

Évolutions attendues et observées (2019-T4 à 2024-T4)

Secteurs Marchands Non-Agricoles / Salariés		Industrie	Construction	Services	Total
Tendance de la productivité par tête / an (%)	(a)	+1,5	-0,0	+0,8	+0,8
Valeur ajoutée (%)					
Évolution cumulée observée de la VA		-1,0	-7,3	+8,1	+5,5
Emploi (%)					
Évolution cumulée attendue	(b-5*a)	-8,0	–7,1	+4,2	+1,5
Évolution cumulée observée	(c)	+3,3	+6,6	+8,4	+6.2
Emplois cumulés à expliquer	(c-b-5*a)	+11,3	+13,7	+4,1	+4,7
Emploi (milliers)					
Évolution cumulée attendue	(d)	–239	–134	+618	+245
Évolution cumulée observée	(e)	+68	+84	+919	+1 431
Emplois surnuméraires à expliquer	**(e-d)**	**+296**	**+259**	**+576**	**+1 186**

INSEE, Ducoudré & Heyer (2017), auteurs.

Note : les valeurs sont arrondies à la deuxième décimale, ce qui peut faire apparaître des écarts significatifs dans ce tableau simplifié, notamment entre les données annuelles et quinquennales en %.

Conclusion

Au total ces modélisations lèvent une grande part du mystère de la productivité en France dans les années récentes. Les contributions des politiques publiques actées dans le cadre du plan de relance, pour certaines prolongées ensuite, expliquent trois quarts des emplois surnuméraires créés depuis 2019 (graphique 6 et 8).

Ces estimations apparaissent stables dans le temps, mais suggèrent aussi en creux que la durabilité de ces emplois est très dépendante des financements publics qui les accompagnent[22]. C'est notamment le cas des deux principaux moteurs qui expliquent 45 % des emplois surnuméraires : l'apprentissage apparaît vulnérable à la recherche d'efficience des dépenses budgétaires tandis que les aides exceptionnelles aux entreprises ont cessé d'être alimentées si bien que leur effet sur l'emploi s'amenuise naturellement. Les contributions positives de la baisse du chômage, du coût et de la durée du travail devraient quant à elles se réguler spontanément sous l'effet du ralentissement conjoncturel en cours.

22. En ceci nos conclusions diffèrent sensiblement de Askenazy *et al.* (2024) qui concluent que les facteurs explicatifs de la panne de productivité qu'ils identifient auraient un effet pérenne sur l'emploi.

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Middle East stasis set to keep Fed on hold at Chair Powell's swan song

Next Wednesday, the Federal Reserve is widely expected to leave monetary policy unchanged in an environment where the economy continues to grow, albeit with intensifying headwinds, and transportation fuel costs are pushing inflation higher

Fed Chair Jerome Powell's term is set to end on 15 May

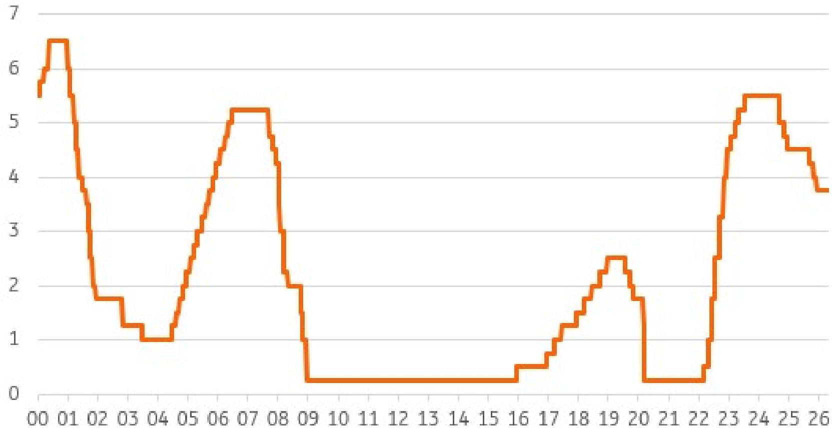
Rate cuts remain more likely than hikes

Absolutely nothing is priced in terms of Fed funds futures contracts for the 29 April Federal Reserve policy decision, but the market still leans in the direction of eventual rate cuts with around 10bp of easing priced by the end of the year. The Fed itself continues to have one 25bp rate cut within its March economic projections for 2026, while the consensus amongst economists remains a little more aggressive, predicting two 25bp cuts in the latter part of the year. We also continue to forecast two rate cuts this year, one in September and one in December.

The main interest will be Jerome Powell's press conference which, in theory, will be his last given his term as Chair ends on 15 May. As such, there will be questions over how he sees his legacy and his relationship with President Trump and whether he will choose to stay on at the Fed given his term as a Governor doesn't end until January 2028. However, it may not actually be his last as Fed Chair. Until Republican Thom Tillis is satisfied that the Department of Justice's "vindictive prosecution" of Jerome Powell, tied to the renovation of the Fed's headquarters, has been brought to an end, the confirmation of Kevin Warsh as his successor will remain blocked at the Senate Banking Committee stage. Powell has already said he will stay on as chair "pro tempore", which would only intensify tensions between the Fed and the President.

In terms of the outlook for monetary policy, we are unlikely to get anything particularly revelatory. The minutes from the last decision suggested that the "vast majority" of the FOMC membership saw employment risks as skewed to the downside, while progress on inflation was likely to be slower because of the economic headwinds being generated by developments in the Middle East. This is likely to remain the way the committee assesses the situation with little prospect of any immediate change in policy signalled.

Fed funds ceiling (%)



Source: Macrobond

Inflation likely to be transitory with the jobs markets still stalled

In terms of inflation, it has been five years since either headline or core consumer price inflation were at or below the 2% target, and higher fuel and airline fares mean CPI looks set to break back above 4% imminently. The Federal Reserve can't do anything to ease the current energy supply shock though. Instead, its focus is to ensure that inflation expectations remain contained and, for now, that seems to be the case. Hence the mildly hawkish rhetoric coming from officials.

Importantly, the current supply shock, focused on fuel prices, is not as broad as the pandemic-related supply chain stresses in 2020/21, and we don't have the same sort of demand impetus that would risk a broader, more persistent inflation story.

Real household disposable incomes are already flatlining, so we see higher fuel prices as being demand destructive via reduced spending power. This is set to weigh on core inflation, and we also need to remember the importance of shelter costs within the US inflation basket (35.5% weighting for headline and 44% for core), which we expect to moderate further. Moreover, if Middle East tensions ease and oil prices drop back to below where they are today, there is a high probability of sub-2% year-on-year inflation being achieved in 2027.

In terms of jobs, there are only 260,000 more people in work today than 12 months ago, implying that the jobs market has effectively stalled during a period when the US growth story was robust. Our concern is that with the Middle East tensions showing little sign of coming to an imminent conclusion, an overlay of heightened geopolitical, economic and market angst is not going to incentivise business to suddenly start hiring now. In fact, with cost pressures rising and consumer spending power being squeezed, corporate profitability will face more challenges and runs the risk that employers will look to cost containment measures. This threatens weaker payrolls numbers in the coming months.

The Federal Reserve has to optimise monetary policy for two very different goals – inflation and jobs. It still views the current stance of monetary policy as being modestly restrictive. With a new, more dovish Fed Chair taking office and key elections this year, which will keep up the political pressure for action, we look for September and December Fed cuts.

The balance sheet and prognosis for Treasuries

Since the T-bills buying programme (re)commenced in mid-December, holdings of bills have risen to over USD$425bn (from USD$195bn). That has expanded the Fed's overall "securities" holdings (including bills) by some USD$185bn. The Fed did this to help boost bank reserves, which had fallen to below USD$3tn, coinciding with repo tightness, and in consequence saw the effective funds rate rise right up to just 1bp short of the rate paid on reserves. Since then, repo pressure has eased, and this week, the New York Fed announced a reduction in monthly T-bills buying from USD$40bn to USD$25bn. This is a purely operational move, completely within its authority and as agreed at a prior FOMC meeting. Still, Chair Powell could get some questions on this, or indeed he could comment on it up front. It certainly suggests a degree of comfort on the part of the Fed over liquidity conditions.

There is also a link here with the incoming chair, Kevin Warsh. He has an ambition to reduce the Fed's holdings of bonds, and in particular, Mortgage Backed Securities. Currently, the Fed holds just under USD$2tn of these, and continues to roll them off as they mature. It's a slow process though, running at some USD$10-15bn per month. Chair Powell could get quizzed on this, too. Any acceleration in roll-off, e.g. outright selling, would place upward pressure on longer-term Treasury yields, as any selling would likely occur in longer tenors. There is no stated plan here. It's all supposition at this stage. But it's very market relevant. The odds are that Chair Powell swats away questions on this, as his FOMC has a steady-as-she goes policy, and he won't comment on what Kevin Warsh might do. But still, worth asking the question(s). Beyond that, the bulk of the Treasury-relevant issues will centre on inflation expectations as seen through the Powell prism.

Potentially some upside risks for the dollar

Even though the Fed story is far less of a driver for the dollar than events in the Middle East, the FX market will still take note of the FOMC statement and Chair Powell's press conference. Part of the market's enduring demand for risk assets this year has rested on the expectation that the Fed will help out with rate cuts (and a weaker dollar).

Depending on how tough Chair Powell sounds on monitoring the inflation threat, and of course, depending on where energy prices are at the time, we see some potential upside risks to the dollar from this FOMC. Should Chair Powell echo any of the recent comments by Christopher Waller that the longer energy prices stay high, the greater chance of unchanged policy, then some bearish flattening of the yield curve could give the dollar a lift across the board.

If that is the case, EUR/USD could be heading down towards the 1.15 area consistent with the 'Prolonged Disruption' scenario presented in our recent update. Depending on how hawkish the Bank of Japan has been at its meeting the day before the FOMC, USD/JPY could be embarking on a sustained push above 160. One week realised USD/JPY volatility has recently dipped to 5%, which is exceptionally low for this pair. Given the unstable global backdrop, April's two-yen range in USD/JPY looks too narrow and at this stage, the upside looks the path of least resistance.

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Bank of England likely to keep rates on hold as markets ramp up hike bets

Financial markets are once again pricing upwards of two Bank of England rate hikes this year. Governor Andrew Bailey won't like that, but with the ongoing crisis in the Middle East, we doubt the Bank will meaningfully push back against market expectations at its 30 April meeting

Bank of England Governor Andrew Bailey speaking earlier this month at the 2026 IMF and World Bank Group meetings

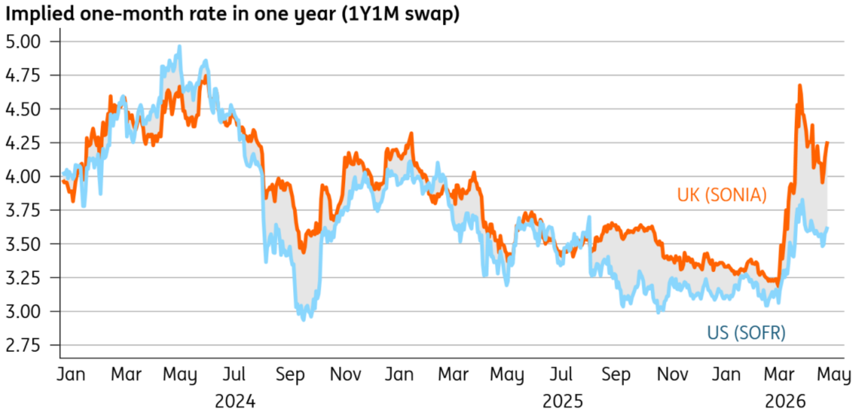
Markets are pricing two rate hikes this year

The Bank of England is poised to keep rates on hold at its April meeting. But will it push back against market expectations of two rate hikes by year-end?

The Bank was evidently caught off guard by the reaction to its March decision, which saw even the most dovish officials appear open-minded about hiking rates. Governor Andrew Bailey told *Reuters* in the days after that investors were "getting ahead of themselves".

We doubt his views have changed since then, but we suspect the Bank will be reluctant to push back too hard on market expectations at Thursday's meeting. For one thing, market pricing is not quite as extreme as it was towards the end of March. That's mostly because oil prices have come lower; the correlation between energy costs and short-dated swap rates has been fairly tight since the crisis began.

UK rate expectations are significantly higher than pre-war



Source: Macrobond, ING

More importantly, though, nobody, least of all the Bank, knows how the crisis will develop between now and June. Energy prices are unpredictable, but the BoE is also acutely aware that the longer the Strait of Hormuz disruption endures, the more likely the unforeseen knock-on effects on supply chains.

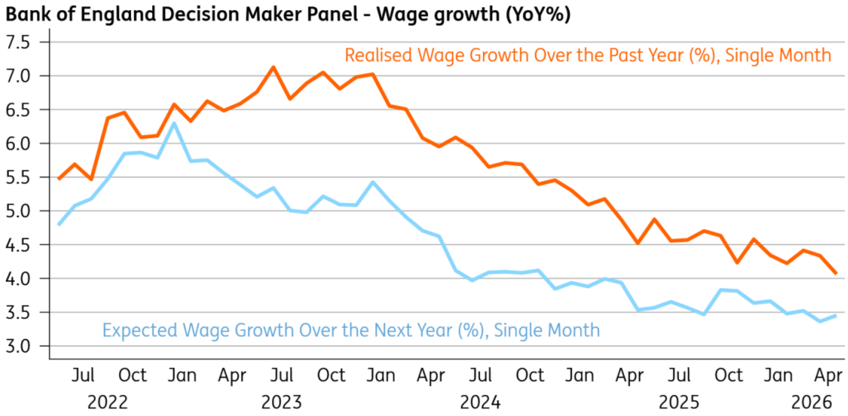
That said, the fact that natural gas prices have stayed remarkably contained – and have even flirted with pre-war levels in recent days – is a significant source of near-term comfort. Natural gas – and the role it plays in setting electricity prices and heating bills – is in many ways a bigger vulnerability for the UK than oil.

The other challenge the Bank has is that the data received since the March meeting hasn't really told it much about how the economy is responding to the war. Inflation is up, but so far only on the predictable rises in motor fuel and heating oil costs. Consumer inflation expectations have also predictably surged.

But measures of corporate price behaviour are more mixed. The PMIs pointed to rising output price pressure. But the Bank's own survey of companies – the Decision Maker Panel – didn't point to an unduly large pick up in inflation expectations. Unsurprisingly, most respondents said the war would lead them to raise prices, but crucially, wage growth expectations – the real driver of longer-term inflation persistence – are unchanged since the crisis began.

In short, it will take time to get a decent sense of whether firms are able to pass on higher energy costs to their consumers. And the Bank will want sight of the April CPI data, which will be released in May. That will give us a sense of how aggressive (or not) price hikes were at the start of the financial year. Given that large swathes of prices in the service sector are only updated once per year, this is fairly consequential for the inflation profile over the next 12 months.

Surveys don't point to a spike in wage growth



Source: Macrobond, ING

Expect an 8-1 decision in favour of "no change"

If the data has told us anything in recent weeks, it is that the economy is in a very different place from 2022 when the last energy crisis broke out. Employment is still falling across the consumer services industries and private sector wage growth has fallen – and crucially is now at a level the Bank told us in February was consistent with achieving the inflation target over the medium-term.

That's why, after last month's surprising show of unity, we expect the old divisions among committee members to come back to the fore. Though the framing has naturally shifted from "cut or hold" to "hold or hike", we'd expect the prior doves to place more emphasis on the differences with 2022 at this meeting than they did at the last.

But we think that's unlikely to extend to formal pushback against rate hike pricing in the policy statement. That's because the hawks are likely to become more vocal, too. We expect at least one official – Chief Economist Huw Pill – to vote for a rate hike. Megan Greene, and less likely, Catherine Mann, might also be tempted to join him. That most likely tees up an 8-1 vote in favour of keeping rates on hold. And at the very least, the hawks will want to keep talking up the possibility of a hike in June in a bid to keep inflation expectations under wraps.

We also doubt we'll get any fresh clues on policy direction from the new forecasts. On current market pricing, we think the Bank will have inflation peaking close to 4% this summer/autumn. But it will likely lean heavily on a scenario approach, given the uncertainty.

Politics is a risk, but we don't expect a rate hike this year

Finally, there's the politics. The Bank's hawkish signals in March were presumably at least in part directed towards lawmakers in Westminster – a warning against any big knee-jerk fiscal intervention in the energy crisis that might necessitate tighter monetary policy. So far, the government has been clear that any support will be targeted – and with natural gas prices where they are, the impetus to do more than that is limited, anyway.

But the risk of a leadership change in Downing Street is growing. Markets are wary that a new prime minister – and, by extension, chancellor – might mean more borrowing and looser fiscal rules.

So in short, we'd expect the Bank to keep its options open next week. Avoid doing anything that could add to rate hike bets in markets, but without trying to actively talk them down, either.

Ultimately, though, we don't think the Bank will hike rates this year – not at the current level of energy prices. We expect rates to stay at 3.75% in April and June, and for the rest of 2026.

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Disclaimer

April ECB cheat sheet: Ready, aim, hold

Markets have two convictions: that the ECB will hold in April, and it will hike in June. We don't expect either of them to be challenged at next week's meeting. The rates market may also have little reason to revise expectations for two hikes in 2026. It's a supporting argument for the euro, but not enough to edge out global equities and oil as the main drivers

Next Thursday, we expect the ECB to stick to a cautious tone while emphasising that both inflation and growth risks remain material

This is our market preview of Thursday's ECB meeting; you can find our macro team's preview here.

The European Central Bank will likely keep rates on hold at the 30 April meeting. That's been hinted at by several Governing Council members lately, who have stressed the lack of new information and the absence of urgency to act. In that sense, April will mainly serve as a reality check for rate expectations, with markets currently pricing around 50bp of tightening by year end.

In the table below, we outline four scenarios from the most dovish to the most hawkish (with the latter including a surprise hike), and what they would mean for rates and the euro. In our baseline, we expect the ECB to stick to a cautious tone while emphasising that both inflation and growth risks remain material. President Christine Lagarde may not give too much away in the press conference, but we expect she'll give enough implicit hints that a summer rate hike is a reasonable expectation.

Four scenarios for the ECB's 30 April meeting

	Inflation outlook	Growth outlook	Interest rates	EUR/USD (1.170)	10Y Bund (3.01%)
Current stance	Short-term upside risks due to war. If persistent, may cause second-round effects	Risks tilted to the downside, especially in the near term, due to the war	Depo rate at 2.00%. Data-dependent, meeting-by-meeting approach		
Dovish	De-escalation reduces upside for inflation and risks of second round effects	Downside risks for growth have increased and may linger after peace	**Hold.** Markets are overestimating tightening. Wide optionality on rates	1.165	2.90%
Neutral ING base-case	Too early for conclusions on inflation implications for policy. Risks still on upside	Balance of risks still to the downside, but no additional concerns relative to March	**Hold.** Cautious but hawkish tone, keeping the door to a June hike clearly open	1.170	3.05%
Hawkish	Upside risks have increased, growing concerns on infl. expectation de-anchoring	Growth risks remain but have not worsened since March	**Hold.** Signalling explicitly one or more hike(s) now likely in the summer	1.175	3.10%
Very hawkish	Inflation impact likely to be bigger than initially thought, requiring immediate action	Despite growth concerns, the focus is firmly on preventing a new inflation crisis	**25bp hike.** Staying data dependent, but signalling risk of more hikes	1.180	3.15%

Source: ING

Rates expect the ECB to act when needed

With just 10% priced in, markets are not expecting the ECB to hike this meeting, but the stakes are higher for June. Over the past weeks, markets have roughly priced in between 20bp and 40bp of hikes by June. Markets are still speculating about the ECB's reaction function, and any hints about a June move will be taken on board.

So far, long-term inflation swaps have remained very stable, reflecting the confidence in the ECB's inflation targeting. Despite large swings in 2Y inflation swaps, the 5Y5Y inflation forward only bumped up from around 2.08% in February to 2.14%, a minor blip by historical standards. To put this into context, the 5Y5Y rose above 2.6% in 2023. To keep market inflation expectations well-anchored, a hike might be needed eventually if oil stays high or rises further.

The big surprise would be the ECB over-delivering compared to market expectations by hiking already at this meeting. The pass-through to longer-dated rates could, however, be limited in this tail risk scenario. If the move were to be packaged in a dovish way, markets would likely interpret the hikes as a front-loading of future tightening and not a radical change in reaction function. In this case, we would see a mild bear flattening of the curve.

If, on the other hand, the hike is complemented with a hawkish narrative, we do see the risk of a perceived policy error. The ECB could be seen as pressing the brakes too hard, which could damage market sentiment, weighing on longer-dated rates.

In a contrasting scenario where the ECB turns overly dovish, dismissing inflation risks as transitory (without actually using the word "transitory"), we might see a significant steepening of the curve. In this case, 2Y rates might edge lower on fewer rate hike expectations, but the increased long-term inflation expectations would push up rates from 5Y and beyond. Again, not our base case, but still a risk to consider.

FX: Surprise needed, or rates stay secondary for EUR/USD

The notion that the ECB is responding more hawkishly than the Federal Reserve to the ongoing energy shock underpinned EUR/USD resilience initially, and then helped fuel the de escalation rally. The two year EUR/USD swap rate differential is now around 20bp tighter than pre war levels, and its mostly positive correlation with oil prices suggests markets see EUR front end rates as more responsive to fresh escalation risks.

That said, while the front end rates beta to EUR/USD has recovered after briefly dropping close to zero in March, global equities are currently a much bigger driver for the pair. Rates, equities and oil are clearly intertwined, but recent sessions have shown that EUR/USD's ability to hold above 1.170 largely depends on risk sentiment staying resilient, even when oil prices face upward pressure.

In other words, it would take a real surprise from the ECB for the impact on EUR/USD to be both material and lasting. Absent that, central bank policy divergence is likely to take a back seat again to oil moves and, above all, global equity performance.

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Kevin Warsh’s Fed confirmation faces tough tests

Kevin Warsh's confirmation hearing to be the next Fed Chair will see him having to tread a fine line between making the case for lower borrowing costs, which helped him get the nomination from the President, and preserving the Fed's inflation-fighting credentials. Politics and the DoJ case against the Fed mean a swift confirmation is unlikely

At his confirmation hearings, Kevin Warsh will have to balance calls for lower borrowing costs and the inflation-fighting credentials of the Federal Reserve

Kevin Warsh's challenge - the President & market credibility

The Federal Reserve has entered its quiet period ahead of the 29 April FOMC meeting, but it will still be drawing headlines given Kevin Warsh's confirmation hearing to become Federal Reserve Chair is being held tomorrow. Once viewed as a hawk when he previously served as a Fed Governor (February 2006-March 2011), he has been nominated by President Trump, who has demanded much lower interest rates from current incumbent, Jerome Powell.

Warsh will inevitably be questioned on how closely aligned he is with the President's thinking. We imagine he will indicate a belief that, over time, lower interest rates make sense. However, preserving market credibility is of paramount importance and there must be justification.

The Iran conflict-induced jump in motor fuel costs creates near-term challenges in an environment where both core and headline consumer price inflation have exceeded the 2% target throughout the past five years. He will be wary of coming across as too dovish. That could heighten market fears of potential unanchored market inflation expectations, prompting higher long-term borrowing costs.

Treasury Secretary Scott Bessent has emphasised that, in his view, the most important metric is the 10Y Treasury yield given its benchmark role and the impact it has on mortgage rates and corporate borrowing costs. He appears to be onboard with a delay to rate cuts, based on recent media comments, but the President is probably not going to be so keen. Nonetheless, if there is a swift conclusion to the conflict and fuel prices drop sharply, Warsh will strongly suggest there is the possibility of his support for rate cuts later this year.

Structurally speaking, Kevin Warsh will emphasise the opportunity for lower borrowing costs over the medium to longer term. He clearly buys into the idea that tech/AI investment will boost productivity and allow the US to grow faster without generating inflation. To be fair, there appears to be support for this view within the broader Fed, given that within the March forecast update, officials revised higher their long-run GDP forecast to 2% from 1.8%, without altering the long-term inflation prediction. The minutes suggested that “several participants… expected higher productivity growth, associated with technological or deregulatory developments, to put downward pressure on inflation."

The Fed’s balance sheet

In addition to the productivity narrative, Kevin Warsh has another theory. Every US$1tn of Fed bond holdings equates to 50bp on the policy rate (approximately). In consequence, balance sheet contraction, through bond roll-offs or outright selling, facilitates rate cuts. This is convenient, as Warsh also believes that the Fed continues to own too many bonds. The build of bond buying was a consequence of past crises, and now that they are no more, much of the bonds should be off-loaded. He’s not been specific about sizes, but said that they should be significantly reduced. And he’s identified Mortgage-Backed Securities (MBS) as a particular eye-sore that should be liquidated. And finally, he’s advocated mostly a passive run-off, cognisant of market de-stabilisation risks.

Consider two scenarios. One is a reduction in the balance sheet right back to pre-Global Financial Crisis levels (5.5% of GDP). That would require some US$4.5tn in bond liquidations. That’s unlikely, but we do develop that narrative here. More probable is a targeted sale of the US$2tn of MBS holdings. This is currently rolling off at a slow trickle, as it's primarily long in duration. But it could be subject to outright sales. It could be softened by extra T-bills buying, to prevent repo market disruption.

The issue here, however, is the impact on bank reserves. Bank reserves sit on the Fed’s balance sheet, and are remunerated by the rate the Fed pays on reserves. They are there as an echo of the prior bond buying undertaken by the Fed (effectively the other side of the Fed’s balance sheet). In 2019 and again through the second half of 2025, we saw repo market tightness occur as bank reserves fell to critical levels. For no particular reason, circa 10% of GDP is a level below which tightness has tended to kick in.

Kevin Warsh has suggested identification of a target for bank reserves, and keeping to that. He’s not been specific as to what that level should be. The unwritten level today is US$3tn. That’s sitting alongside Fed bond holdings of US$6.3tn. Simplistically, if the Fed’s balance sheet was shrunk by

US$2tn, bank reserves could be pressured down to around US$2-2.5tn. In today's setup, that would cause repo issues.

The simple solution would be to offset MBS sales with T-bills buying. Not technically QE (as bills buying is short in maturity and, in a sense, akin to open market operations), but would still leave the balance sheet bloated with government securities. Warsh needs a plan to deal with this. We'll be listening for more as he testifies, albeit perhaps a bit deep in the weeds for a Senate hearing. It is *really* important though. Apart from repo issues, MBS selling with a view to cutting rates, points to a steeper curve from both ends.

Chair Powell could continue beyond 15 May

Whether Kevin Warsh will be approved quickly is another matter given the Department of Justice's probe into Fed construction work overspending. The Senate Banking Committee consists of 13 Republicans and 11 Democrats, but Republican Thom Tillis has vowed to block the confirmation until the "vindictive prosecution" of Chair Powell is ended. Until Thom Tillis is satisfied, it will be a 12-12 vote, and it can't proceed to a formal Senate vote.

Jerome Powell's term as Fed Chair notionally ends on 15 May, but until Warsh is approved, Powell will stay in the role with his term as Governor scheduled to continue until 31 January 2028. As such, tensions between the Fed and the President are likely to remain elevated, but with strong rules of governance, we think the Fed's independence is not in doubt

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